

Geospatial MCE of Residential Livability and Hydrological Vulnerability in Xinzhuang District

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1. INTRODUCTION

1.1 Research Motivation and Urban Development Background

As global urbanization accelerates and Transit-Oriented Development (TOD) continues to advance, the paradigm of modern urban planning has shifted from traditional expansion-oriented approaches to a Smart Growth model that prioritizes residents' Quality of Life (QoL) and environmental sustainability. As a core development area on the western side of the Greater Taipei Metropolitan Area, Xinzhuang District in New Taipei City has become a hotspot for population influx and spatial redevelopment in northern Taiwan, having undergone high-intensity commercial development in the Xinzhuang Sub-city Center and the transformation of the urban landscape in the Xintai Wenzaijun Redevelopment Zone. Faced with a large influx of potential homebuyers, evaluating the residential livability of a locality can no longer rely solely on market housing prices from an economic perspective. A more pragmatic approach must entail a comprehensive assessment from a geospatial dimension, which encompasses residents' daily accessibility to ecological green spaces, the connection efficiency of active green

transportation, the service functions of public facilities, and the traffic safety and environmental quality disruptions around neighboring community roads.

1.2 Spatial Characteristics and Urban Spatial Conflicts of the Xinzhuang Watershed

The spatial characteristics of Xinzhuang District are complex due to its unique hydrological network. The five major tributaries on the left bank of the Dahan River—the Zhonggang Drainage Channel, Taliaokeng Creek, Tandigou, Shibafernkeng Creek, and Yakoukeng Creek—geographically span across different land use zones, which in turn gives rise to three distinctly different urban spatial issues:

First is the urban waterfront space, represented by the remediated Zhonggang Drainage Channel, which has now been transformed into a high-quality urban Blue-Space Infrastructure, providing a significant environmental premium and recreational functions for residential areas on both sides. In contrast, there are the vulnerable environmental quality zones left behind by early industrial development, primarily located along Tandigou and the downstream reaches of Taliaokeng Creek. These channels flow through traditional, densely

populated industrial areas and face long-term challenges of channel siltation and industrial wastewater interference; certain channel sections even experience severe odor issues during specific seasons, acting as a persistent environmental disamenity for surrounding residences.

The aforementioned "Spatial Overlay" of land use and hydrological networks has led to a significant spatial conflict between residential livability and environmental interference in Xinzhuang District. Although some early-developed old urban areas offer exceptionally high convenience regarding commercial activities and public facilities, they are often immediately adjacent to poorly remediated drainage trunks or high-volume commuter arteries, exposing local residents to elevated environmental and traffic risks.

1.3 Research Concepts and Multi-Criteria Spatial Analysis Framework

To quantitatively evaluate the aforementioned spatial conflicts, this study introduces the concept of the "15-Minute City" (Moreno et al., 2021), which has been widely promoted in urban planning circles in recent years. This theory emphasizes pedestrian-oriented transportation, arguing that residents

should be able to meet their basic daily needs—such as shopping, healthcare, education, and recreation—within a 15-minute walk or bicycle ride (translated into an actual service radius of approximately 1,000 meters based on an adult walking speed of 1.2 m/s).

The multi-criteria decision-making sub-models constructed in this study encompass the following five core spatial analysis dimensions:

1. Multi-ring buffer analysis of urban green spaces and watershed corridors.
2. Isochrone analysis of active transportation networks and multi-modal public transit transfer accessibility.
3. Multi-conditional overlay operations of public facility life circles under vector road networks.
4. Kernel Density Estimation (KDE) of road traffic accident hotspots.
5. Exploration of human activity levels within virtual activity landscapes utilizing Volunteered Geographic Information (VGI).

Ultimately, this study will utilize the Multi-Criteria Evaluation (MCE) overlay analysis method within Geographic Information Systems (GIS) to perform grid-

level weighted overlay operations on these heterogeneous layers. Through the generated Composite Livability Index, this framework eliminates the blind spots of past subjective site selection, providing a more scientifically proven spatial decision support system for private home-buying decisions and urban planning authorities.

2. STUDY AREA

2.1 Geographical Location and Administrative Boundary

Definition

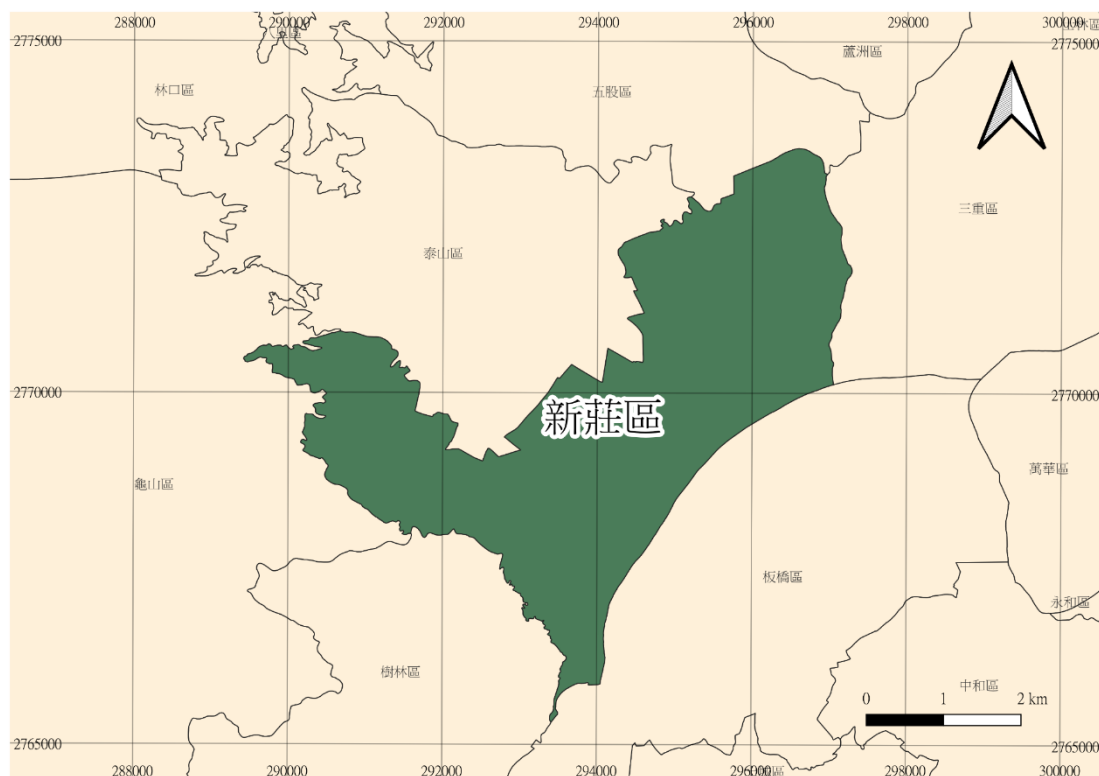


Figure 2.1 Regional Map of Xinzhuang

The geographical scope of this study focuses on the entire area of Xinzhuang District, New Taipei City, Republic of China (Figure 2.1). Located on the western edge of the Taipei Basin, Xinzhuang District is bordered by Taishan District and Wugu District to the north, faces Banqiao District and Sanchong District across the Dahan River to the east, connects with Shulin District to the

south, and adjoins Guishan District of Taoyuan City to the west along the boundary of the Linkou Plateau. The total area of the district is approximately 19.74 square kilometers, and its administrative divisions encompass 84 villages (Li) and 1,836 neighborhoods (Lin). According to the latest population statistics from the Department of Household Registration, Ministry of the Interior, the resident population of Xinzhuang District has reached 420,000, with a population density exceeding 21,303 people per square kilometer. This places it among the highest-ranked townships, cities, and districts in Taiwan, characterizing it as a typically high-population-density and highly developed urban unit within the Greater Taipei Metropolitan Area.

2.2 Topographical, Geological, and Land Use Characteristics

In terms of topographer and geomorphic units, Xinzhuang District primarily spans two distinctly different geomorphic structures: the eastern and southern portions belong to the Dahan River alluvial plain, which features open and flat terrain with an average elevation between 2 meters and 10 meters, serving as the core zone for early industrial, commercial, and dense older residential development; the western and northwestern portions gradually transition into the foothill fringes and slopes of the Linkou Plateau (such as Danfeng and

Huilong). The contemporary land use characteristics of Xinzhuang District exhibit a high degree of Spatial Heterogeneity, a phenomenon deeply rooted in the evolution of its urban morphology over more than two centuries.

Situated at the critical geographical confluence of the Dahan River and the Tamsui River, Xinzhuang was part of the core area in the early development of the Greater Taipei Basin.

During the Qianlong and Jiaqing eras of the Qing Dynasty, Xinzhuang developed into a typical river port settlement by virtue of its water transport advantages along the Dahan River. However, after the mid-Jiaqing period, severe channel siltation and port abandonment occurred along the Tamsui River channel (geographically close to the contemporary location of the Erchong Floodway), causing its shipping functions to decline sharply. Consequently, the core of urban development underwent a spatial shift, and the district temporarily stagnated.

It was not until the late Japanese colonial period that the Governor-General of Taiwan, in response to the risk of Allied air raids during World War II, promoted an industrial decentralization policy that forced some core factories in Taipei City to disperse to surrounding townships. In the early post-war

period, the government continued this industrial trajectory and actively promoted expanded industrialization, leading to the low-cost expansion of the "Touqian Industrial Zone" and "Xisheng Industrial Zone" in succession.

Xinzhuang thus underwent a drastic transformation from a traditional agricultural settlement into a highly industrialized urban unit characterized by manufacturing clusters. This historical context directly shaped the rigid landscape features of mixed residential-industrial land use across older urban districts in Xinzhuang.

This historical development of mixed residential-industrial land use directly resulted in the contemporary north-south spatial divergence of land use in Xinzhuang:

The Emerging Integrated Development Zones of North Xinzhuang: This includes the Xinzhuang Sub-city Center and the Touqian Redevelopment Zone—both key development focuses in recent years—as well as the Xintai Wensing Redevelopment Zone, which has undergone large-scale landscape transformation recently. Benefiting from modern urban planning guidance, the road system in this area is mostly designed in a grid pattern. The allocation of public facilities is relatively complete, featuring a high proportion of statutory

green spaces, commercial and administrative land uses such as the Joint Office Tower, and modernized high-rise exclusively residential areas.

South Xinzhuang and the Linear Older Urban Districts: Developed primarily along both sides of Zhongzheng Road, Xinshu Road, and Huacheng Road, this area is heavily impacted by the historical fabric of early mixed residential-industrial land use, resulting in a dense distribution of small-and-medium-sized chemical, textile, machinery, and corrugated metal sheet sheet-metal factories. The buildings here mostly consist of low-rise townhouses and traditional apartments. Due to the lack of forward-looking regulations in the early stages, urban public facilities appear relatively constrained, and the high density of rigid structures and industrial workshops has made the phenomenon of highly impervious surfaces on the ground level extremely prominent.

2.3 Watershed Network and Hydrological Spatial Zonation

In addition to the north-south divergence in land use, the most critical environmental and geographical feature of Xinzhuang District lies in the hydrological network system on the left bank of the Dahan River. Under the dual influence of topographical barriers and intense urban development, the

five core tributaries within the district—the Zhonggang Drainage Channel, Tandigou, Taliaokeng Creek, Shibafernkeng Creek, and Yakoukeng Creek—exhibit distinct environmental landscape benefits alongside negative quality interference characteristics:

First is the Zhonggang Drainage Channel, which flows through the core of North Xinzhuang and serves as the primary urban drainage artery within the Xinzhuang Sub-city Center and the Touqian Redevelopment Zone. Historically, due to receiving massive volumes of upstream manufacturing wastewater, its water quality remained severely polluted for a long period, making it a persistent chronic ailment for local environmental governance. Following urban blue-ribbon landscape remediation, it has now successfully transformed into a multi-functional hydraulic space that combines urban flood retention adjustment with a high-quality waterfront corridor (Blue-Space Infrastructure). Lined with high-end residential properties on both sides, it serves as a core indicator for the ecological livability bonus evaluated in this study.

Second is Tandigou, located within the Xisheng and Min'an life circles of South Xinzhuang. Hydrologically, it constitutes a vital distributary of Taliaokeng Creek, which itself is a tributary of the Dahan River. Dense traditional small-

and-medium-sized industrial zones are distributed along both sides of the Tandigou watershed, presenting long-standing spatial challenges such as severe channel siltation, surroundings dominated by highly impervious surfaces, and poor industrial water quality.

Finally, there are the upstream reaches of Taliaokeng Creek, Shibafernkeng Creek, and Yakoukeng Creek, distributed across the western portion of Xinzhuang near the foothill fringes bordering Shulin District and Guishan District. Originating from the edge of the Linkou Plateau and flowing down toward the alluvial plain, these tributaries create spatial overlays with multiple major transportation arteries, including Zhongzheng Road and the MRT Xinzhuang Line rail corridor. Because these three water systems currently maintain mostly natural river channel patterns or remain partially constrained by narrow natural topography, their surroundings lack waterfront recreational facilities plan like that of the Zhonggang Drainage Channel. Furthermore, being immediately adjacent to transportation arteries, they present spatial characteristics of low waterfront utility and significant functional isolation.

In summary, Xinzhuang District brings together spatial characteristics featuring overlapping historical industrial and commercial fabrics, high-

intensity development within redevelopment zones, and the environmental quality differentiation of blue and green ribbons. Through a geospatial analysis of this district, an empirical model balancing environmental quality, traffic safety, and residential livability within a high-density urban environment can be effectively provided.

3. DATA COLLECTION

3.1 Data Collection and Criteria Standards for Urban Green

Space and Environmental Justice Analysis

This study achieves a highly refined integration of supply-side spatial geometry and demand-side resident population distribution data for the spatial accessibility and environmental justice analysis of urban planning public facilities (including parks, plazas, and stadiums) in Xinzhuang District. Regarding the construction of the supply-side spatial layers, the research team extracted the *Urban Planning Land Use Zoning and Boundary Map of New Taipei City* issued by the National Land Management Agency, Ministry of the Interior. Through structured querying within the attribute tables of the Geographic Information System, geometry data for pure green space assets designated as parks, green spaces, children's playgrounds, plazas, stadiums, and multi-functional flood retention ponds within the Xinzhuang Urban Planning Area were precisely isolated to serve as the baseline supply for spatial analysis.

On the demand-side resident population base map, to thoroughly avoid the spatial ecological fallacy caused by uneven population distribution within

administrative boundaries when using traditional administrative villages (Li) as statistical units, this study introduces the "Minimum Statistical Units (MSU) Map Data" from the National Geospatial Data Common Standards. Cross-platform integration was performed with the *Resident Population Statistics of New Taipei City Statistical Districts* from the Statistical Information Service of the Ministry of the Interior (SEGIS). The entire district comprehensively covers 1,334 minimum street block units, aiming to precisely display the spatial clustering characteristics of the population at a micro-scale.

Regarding the evaluation regulations and data standards for spatial suitability, this study primarily relies on the statutory public facility ratio standards in Article 45 of the *Urban Planning Act*, which stipulates that the total area of parks, green spaces, plazas, stadiums, and children's playgrounds designated within an urban planning area shall, in principle, not be less than 10% of the total planning area. At the local policy execution level, this corresponds to the greening and beautification performance indicators from the *New Taipei Green Home Project* by the Department of Land Administration of the New Taipei City Government and the National Development Council. For the threshold setting of per capita Quality of Life (QoL), referencing the average

per capita park and green space area within planned areas of major metropolitan regions (such as Taipei City and New Taipei City) in the urban planning statistics of the National Land Management Agency, Ministry of the Interior (which ranges between approximately 3 and 5 square meters), this study sets the major domestic metropolitan average of 4.0 square meters per person as the spatial threshold for evaluating whether a street block achieves excellent livability. Concurrently, the study further introduces the basic healthy urban life standard of 9.0 square meters per person recommended by the World Health Organization (WHO) as an environmental justice benchmark to measure whether the emerging redevelopment zones of North Xinzhuang and the older urban areas of South Xinzhuang meet international top-tier livability reference points.

Furthermore, considering the correlation between site green coverage and the Urban Heat Island (UHI) effect, Chapter 7 "Green Buildings" of the Building Design and Construction volume of the current *Building Technical Regulations* and the *Regulations for Greenery Implementation in New Buildings* promulgated by local governments require that the green coverage rate of new construction sites must reach a statutory standard of 50% or higher based

on different land use zones. However, as described in the development context of Chapter 2, the early-developed older urban life circles of South Xinzhuang, characterized by a mix of factories and dense residential areas, lacked such statutory regulations for public facilities and green coverage during their early development years, thereby forming a vacant zone devoid of public green spaces. This study incorporates this historical background into the multi-criteria spatial indicators to demonstrate how the high heat-retention climate risks triggered by high-density impervious surfaces and a lack of green coverage intertwine with the resident population to create an environmental injustice that urgently requires urban renewal to reverse.

Finally, regarding the geographical indicators set for spatio-temporal accessibility and road detour ratios, based on an average adult walking speed of 1.2 m/s, the theoretical straight-line moving distance for a citizen walking for 15 minutes is approximately 1,080 meters. To align with the road detour normality and network geometry science of Geographic Information System (GIS) spatial analysis, this study adopts a 1,000-meter service radius (Buffer) commonly used in urban planning academia as the basic fulfillment threshold for "spatial accessibility," and sets a 500-meter service radius—representing

approximately a 7-minute walk—as the optimal standard for a "high-quality green life circle." For water environments, the research team also collected vector layers for park points and water bodies such as the Zhonggang Drainage Channel, Tandigou, and Taliaokeng Creek within Xinzhuang District, establishing positive bonus buffers based on the aforementioned 15-minute walking accessibility. Additionally, a 200-meter proximity impact negative environmental interference buffer was established for the industrial sections of Tandigou, which have a history of pollution, odor, and flooding, serving as a spatial heterogeneity indicator for subsequent livability deductions in the MCE model.

3.2 Data and Standards for Bicycle Networks and Multi-Modal Public Transit Stations

To quantify the accessibility of the active and green transportation network in Xinzhuang District, the research team comprehensively selected spatial vector data from official government open data platforms as the analytical baseline. Regarding the transportation network and public transit stations, spatial point data within Xinzhuang District and its surrounding buffer zones were completely extracted. This includes the "MRT Station Spatial Data" provided by

the Government Open Data Platform (used to define the core nodes of Transit-Oriented Development, TOD), as well as "Bus Stop Information" and "New Taipei City Public Bicycle Rental System (YouBike 2.0) Station Data" provided by the New Taipei City Government Open Data Platform—the latter two serving as critical bases for evaluating micro-mobility and last-mile transfer analysis. Furthermore, to empirically analyze the continuity and reachability of the active transportation network, the "New Taipei City Bicycle Lane Data" was imported to locate the access points of bicycle lanes. In terms of hydrological spatial data, to quantify the spatial distribution of the five core water systems within the district, the latest version of the "River Basin Boundary Map" and linear vector data of the water systems were acquired through the Water Resources Spatial Information Service Platform of the Ministry of Economic Affairs, thereby defining the potential high-risk inundation zones of low-flood-resilience watersheds.

To convert these diverse, heterogeneous datasets into comparable geographical indicators, this study establishes spatial analysis and accessibility standards based on the station planning principles and facility requirement specifications of the Department of Rapid Transit Systems, New Taipei City

Government. Regarding the accessibility settings for MRT stations, bicycle rental stations, and bus stops, considering that average passengers are typically willing to walk approximately 10 minutes for transfers, the corresponding theoretical service radius and walking distance fall between 500 and 700 meters. In view of this, this study takes the inner core boundary and uniformly adopts 600 meters as the baseline for transport accessibility and multi-ring buffer analysis to measure the service efficiency between residential properties and public transit nodes.

However, concerning the accessibility standards for the bicycle lane access points along the banks of the Dahan River, this study does not reuse the 600-meter indicator applied to MRT stations, but instead adjusts it to a 900-meter buffer radius based on two academic and practical considerations:

First, there is a fundamental difference in spatial positioning and movement motivation. MRT stations, bus stops, and bicycle rental stations primarily accommodate high-frequency, compulsory daily commuting, where residents' spatio-temporal tolerance for walking is usually strictly limited to within 10 minutes (600 meters). In contrast, bicycle lane access points are mostly located along waterfronts or within specific riverside green belts outside the dikes,

serving as recreational or specialized transfer nodes. Users at these nodes generally possess higher recreational motivations and are willing to expend greater spatio-temporal costs, extending their journey to the 15-minute walking limit (approximately 900 meters).

Second, there is a substantial benefit generated by micro-mobility vehicles during the last mile. Although 900 meters is close to a 15-minute walk, it actually takes only 3 to 4 minutes of riding time if residents ride a YouBike 2.0 to get there. Therefore, adopting 900 meters as the service envelope for bicycle lane access points can more reasonably and completely encompass the substantive service coverage of the active transportation network over surrounding urban communities, preventing an underestimation of the extended synergistic effects of green transportation.

3.3 Point Data for Urban Public Facility Convenience

To comprehensively deconstruct the spatial functions and suitability of the "15-Minute City" life circle in Xinzhuang District, this study integrates supply-side and demand-side data regarding critical public facilities that influence residents' daily walking life functions. In building the layers, the research team

adopted a "Heterogeneous Spatial Data Fusion" strategy, performing a spatial multi-conditional geometric union (Spatial Layer Merge) between official open data from government agencies and crowdsourced spatial databases from OpenStreetMap (OSM). This approach balances the statutory authority of official statistics with the real-time responsiveness of civic data, precisely constructing spatial indicators across the following five facility dimensions:

First are daily shopping nodes. This indicator evaluates the foundational convenience for residents obtaining daily consumer goods. In terms of layer integration, considering the spatial distribution differences between traditional retail and modern commercial chains, the core points for "traditional public retail markets" and "supermarket chains such as PX Mart and Simple Mart" were downloaded in their entirety from the official datasets of the New Taipei City Government. Concurrently, the spatial geometries of "convenience store chains," which citizens rely on with high frequency, and high-end, large-volume "hypermarkets and lifestyle department stores such as Carrefour and IKEA" were directly filtered and extracted from the OSM international open geographic database. These heterogeneous point layers were ultimately

combined through a spatial geometric union to map the basic material acquisition network of Xinzhuang District.

Second are life support nodes. This indicator centers on infrastructure that maintains daily social and economic operations while promoting the physical and mental well-being of citizens. To ensure the accuracy of spatial positioning, the research team extracted large landmarks within Xinzhuang District, such as the "Xinzhuang Civil Sports Center," as well as branches of major domestic banks and post offices including standard ATM locations that sustain economic activities, from the OSM international open geographic database. Simultaneously, the layer incorporates a geometric union of spatial data for local government agencies that act as grassroots social and administrative support networks, including village community centers, the Xinzhuang District Office, and household registration offices, reorganizing them into a logistical support indicator for urban life.

Third are childcare and cultural nodes. To comprehensively evaluate the accessibility of cultural and educational resources from pre-school to primary school levels across various street blocks in Xinzhuang, this layer merges the official "Directory of Elementary Schools" from the Education Department of

the New Taipei City Government with datasets on public and private kindergartens and daycare centers from the Social Welfare Department. This is supplemented by point data for public library branches and local arts and cultural exhibition spaces recorded in the OSM spatial database, thereby quantifying the spatial coverage of cultural, educational, and childcare resources.

Fourth are MRT commuter transit nodes. This indicator focuses on the core supply of rail transit, serving as the foundation for Transit-Oriented Development (TOD). This layer precisely extracts and integrates the physical entrance and exit geometric points of existing MRT stations within Xinzhuang District (including the Zhonghe-Xinlu Line, Circular Line, and Airport Line), serving as the baseline source for subsequent calculations of rail transit spatial clustering and radiation density.

Fifth are urban safety nodes. This indicator represents a critical line of defense for measuring basic medical resilience and public safety guarantees in urban life. The research team utilized the OSM spatial database to filter out medical resource points within Xinzhuang District, including regional general hospitals (such as Fu Jen Catholic University Hospital and Xintai Hospital), primary

medical clinics, and National Health Insurance contracted pharmacies.

Simultaneously, the layer integrates the spatial geometries of police departments, local police stations, and fire stations responsible for local public security and disaster prevention operations, aiming to quantify the service radius of each street block grid in Xinzhuang concerning emergency disaster relief and medical safety.

3.4 Road Traffic Accident Risk Data

In multi-criteria decision-making models for urban residential site selection and livability, evaluating negative constraints and risks in the urban environment is just as essential as assessing positive public facility supply and convenience to serve as a weighting factor for housing safety. Among numerous urban living risks, the road traffic environment most directly impacts citizens' personal safety and daily walking quality.

To precisely identify road safety threats within Xinzhuang District, this study retrieved the latest 2025 (Year 114 of the Republic of China) "Traffic Accident Data Involving Injuries or Fatalities" documented by the National Police Agency from the Government Open Data Platform. Through preprocessing in

a Geographic Information System, the research team filtered and extracted physical location point data for all "Category A2 traffic accidents" within the administrative boundaries of Xinzhuang District. Category A2 traffic accidents are officially defined as "traffic accidents that result in injury, or death occurring more than 24 hours after the accident."

3.5 Supplementary Indicator—Volunteered Geographic Information and Human Activity Level Data

Traditional urban planning often leans heavily on rigid statistical data issued by official government agencies when evaluating residential livability, which easily overlooks spontaneously generated secondary leisure vitality and informal neighborhood pedestrian activity. To complement this research perspective, this study innovatively introduces "Volunteered Geographic Information" (VGI) as a supplementary indicator.

Utilizing third-party spatial extraction tools, the research team completely exported the spatial vector point data of virtual stations, big flowers, and regular player gathering nodes from the augmented reality (AR) mobile game *Pikmin Bloom* across the entire area of Xinzhuang District. Because the

deployment logic of this game's virtual landscape relies heavily on physical nodes in the real world that possess high pedestrian mobility, pedestrian-friendly environments, and local community vitality, this study treats the clustering intensity of these crowdsourced virtual activity nodes as an emerging indicator of Digital Footprint Activity. By analyzing the spatial clustering characteristics of these virtual activity nodes, this study aims to explore whether a deep spatial coupling relationship exists between virtual activity levels, physical pedestrian walkway safety, and community functions, thereby providing homebuyers with a more diverse microscopic indicator of residential environmental suitability that carries human-centric pedestrian reference value.

4. METHODS

4.1 Spatial Analysis Methods of Per Capita Green Space

The spatial econometrics regarding the environmental justice dimension in this study are primarily conducted based on the QGIS software platform. The core implementation lies in measuring the substantive per capita green space indicator for the resident population within each street block across Xinzhuang through topological operations of pure vector geometry, serving as environmental supply-side data for subsequent residential livability assessments.

The specific operational procedures are strictly confined to the vector data model and are divided into two core steps:

The first step involves vector geometric intersection and clipping. We perform a spatial Intersection operation between the geometric layer of public facility green spaces from the Xinzhuang urban plan and the base map of the minimum statistical units of the resident population, which completely covers 1,334 street block units. This step aims to precisely fragment and attribute discrete park and green space geometric areas based on the administrative

spatial boundaries of the Minimum Statistical Units (MSUs), ensuring that the supply side (green space area) and the demand side (street block population) are perfectly aligned on spatial units.

The second step involves attribute field econometrics and mathematical constraints. Upon completing the geometric intersection, this study utilizes the Field Calculator within the QGIS attribute table to establish data indicators. To force the exclusion of vacant street blocks that have a resident population count (P_CNT) of zero due to being industrial zones, purely commercial districts, or designated for specific public facilities when calculating the per capita green space area for each micro-street block, and thereby avoid system errors (NULL values) triggered mathematically by dividing by zero, this study codes the following conditional constraint formula into the Field Calculator:

```
WHEN "P_CNT" IS NULL OR "P_CNT" = 0 THEN "Grn_Area"
```

```
ELSE "Grn_Area" / "P_CNT"
```

Through the filtering and execution of this formula, the system smoothly generates the critical environmental justice assessment field Per Green (i.e., the per capita park and green space area within the minimum statistical unit)

in the final analytical polygon layer, preserving the geometric accuracy of the vector boundaries of the minimum statistical units to serve as the empirical foundation for subsequent analyses of environmental resource inequality.

4.2 Isochrone Analysis and Gradient Proximity Analysis for Active Transportation

The overall research methodology for the transportation accessibility analysis in this study is divided into data preprocessing and three core modules:

Data Preprocessing—Spatial Scope Clipping and Boundary Data Retention
(Data Preprocessing & Boundary Management)

To ensure spatial data consistency, this study imports the baseline information layers of the entirety of New Taipei City and utilizes the administrative boundary of Xinzhuang District as the primary spatial mask. A Clip spatial operation is executed to restrict vector data—such as number of YouBike 2.0 docks, bus stops, MRT entrances/exits, and river networks—within the administrative boundaries of Xinzhuang. However, considering that bicycle lane access points are mostly located on the outer fringes of the Xinzhuang administrative boundary (such as the riverside green belts outside the dikes),

direct clipping would lead to the loss of critical transportation nodes near the border, failing to realistically reflect residents' cross-border movements into active transportation corridors. Therefore, this study deliberately retains the boundary-adjacent external points for the bicycle lane access points layer (excluding them from the clipping process) to comprehensively evaluate the accessibility of substantive transportation resources on the periphery of Xinzhuang.

First Module: Spatial Join and Counting of Multi-Modal Transit Resources (Multi-Modal Transit Density Analysis)

Using the minimum statistical units of Xinzhuang District as the analytical units, point data for YouBike 2.0 Station Count, MRT station entrances/exits, and bus stops are aggregated through Spatial Join and Point-in-Polygon Counting methods to calculate the total number of bicycle docks, MRT entrances/exits, and bus stops within each minimum statistical unit.

Subsequently, based on the distribution density of each facility type, the choropleth mapping method is applied using gradient colors to illustrate the supply level of active transportation modes and the service intensity of public transit across different areas. This further analyzes the spatial distribution

characteristics and accessibility variances of low-carbon transportation resources throughout Xinzhuang District.

Second Module: Composite Buffer Construction and Multi-Stage Buffer

Intersection Area Operations (Buffer & Intersection Analysis) To quantify the physical spatial accessibility and multi-modal transfer coverage of public transit services, this study performs spatial operations on buffer boundaries and intersections. First, circular buffers with a radius of 600 meters are generated around the original location points of YouBike 2.0 stations, bus stops, and MRT entrances/exits to visually examine the spatial boundaries of a single transit mode's service coverage. Next, to compute the comprehensive coverage rate of the composite public transit network, this study executes a two-stage spatial Intersection overlay operation:

- **Stage One (Dual-Transit Composite Coverage Area):** An intersection operation is performed between the YouBike 2.0 600-meter buffer and the bus stop 600-meter buffer to isolate the scope simultaneously covered by both, representing the "dual service network of basic active transportation and highway passenger transit," which is displayed in light purple on the map. This method leverages spatial geometric

operations and invokes the geometry measurement functions of the Field Calculator to extract the independent geometric areas of this intersection layer and the Xinzhuang plain area respectively, serving as the computational basis for subsequently evaluating the coverage rate of the foundational transfer network.

- **Stage Two (Triple-Transit Core Coverage Area):** The results of the aforementioned dual-transit intersection are subjected to a secondary intersection operation with the 600-meter buffer zones of MRT stations, delineating the highly efficient, multi-modal transport core network where "MRT, bus, and YouBike operate as a trinity," which is visualized on the map. At this stage, the study similarly utilizes GIS tools to capture the geometric area of this triple-intersection polygon layer and calculates its proportion relative to the total area of Xinzhuang District, aiming to quantify the physical spatial coverage rate of the triple-transit core network.

Third Module: Spatial Relationship Analysis of Bicycle and Waterfront

Corridors (Interdisciplinary Spatial Corridors) Finally, this study overlays the bicycle lane access points—which were left unclipped to fully preserve

peripheral nodes—with the physical river network layer of Xinzhuang, integrating them with the aforementioned public transit point layers. This step aims to explore whether the internal commuter-oriented active transportation system of Xinzhuang District (the YouBike, primary bus, and MRT networks) physically intertwines with the external blue-ribbon waterfront corridors (such as the banks along the Dahan River). Because the entrance and exit points on the outer fringes of Xinzhuang are completely preserved, this study can more accurately evaluate how Xinzhuang residents seamlessly transition from the district's internal commuter network to external recreational active waterfront spaces via boundary nodes, thereby measuring the substantive spatial threshold for the integration of public transit and recreational green ribbons.

4.3 Multi-Conditional Intersection and Union Operations for Public Facilities

To transform discrete public facility points into a spatial distribution with geographic context, this study employs a "Grid-Based Multi-Dimensional Density Analysis" and "Composite Service Buffer Overlay Geometric Operations." The overall algorithm and analytical workflow are as follows:

The First Step: High-Resolution Spatial Grid Generation Based on the overall boundary of Xinzhuang District, a 100x100 (or as practically configured) spatial vector grid is constructed to serve as the unified basic statistical unit (Spatial Sampling Unit) for all layers.

The Second Step: Itemized and Consolidated Grid-Based Density Analysis

Utilizing Spatial Join and the Field Calculator, we calculate the number of facility points contained within each independent 100x100 grid unit to derive the spatial distribution density in terms of points per square kilometer. In this step, this study independently generates itemized grid density layers for the five major dimensions: daily shopping, life support, childcare and culture, MRT commuting, and urban safety. Upon completing these itemized grid statistics, these five layers are combined via a spatial Union and weighted summation to produce a consolidated comprehensive density map (Total_Density_Grid) that represents the global clustering characteristics of all facilities.

The Third Step: Service Buffer and Administrative Base Map Intersection

Operation In accordance with international spatial planning standards for the "15-Minute City," this study establishes a uniform circular spatial buffer with a radius of 900 meters around all facility locations within the consolidated

comprehensive functionality layer. Finally, using the Intersection tool within vector geoprocessing utilities, this 900-meter composite buffer layer is overlaid and clipped with the Xinzhuang District administrative boundaries and physical road base maps. This effectively delineates composite life circle functional hotspots and accessibility scopes that align with actual urban boundaries and possess spatial continuity.

4.4 Spatial Econometric Methods for Road Traffic Accidents

4.4.1. Coordinate Transformation

To ensure that "meters (m)" can be used as the precise, statutory unit of measurement during subsequent spatial geometric and area topological calculations, the raw layer in the spherical latitude and longitude coordinate system received from the open data platform was uniformly converted to the Taiwan Island 2-Degree Transverse Mercator Projection Coordinate System (TWD97 / TM2 Zone 121, or EPSG:3826), thereby guaranteeing the precision of spatial displacements and geometric measurements.

4.4.2. Construction of Road Network Micro-Buffers

Using the administrative boundary of Xinzhuang District as a geometric template, a Clip operation was performed on both accident points and road lines to ensure data purification. Considering that law enforcement personnel encounter routine spatial errors when performing GPS positioning at accident scenes, a direct intersection between "infinitely thin vector lines" and "points" would result in a substantial undercount. To address this, this study generated a micro-buffer extending 10 meters on each side of the clipped Xinzhuang road line layer. These elongated spatial polygons accommodate GPS errors while remaining precisely confined within actual physical road widths.

4.4.3. Linear Spatial Intersection and Segment Counting

Utilizing the "Count Points in Polygon" utility within QGIS, the purified accident points were snapped and aggregated into each road micro-buffer to calculate the cumulative total number of traffic accidents on each physical road segment.

4.4.4. Jenks Natural Breaks Optimization

To scientifically visualize the calculated continuous street block accident densities, the Jenks Natural Breaks Optimization method was applied to

classify the entire district's road network into 7 traffic risk levels. To effectively guide visual focus, a "Multiply" blending mode was applied to the spatially classified layers, and minor alleys with zero or negligible accidents were rendered transparent. This map design achieves a high-contrast effect where major arteries appear to glow according to their risk level, while minor alleys seamlessly recede into a clean base map.

4.5 Spatial Clustering Analysis Methods for Volunteered Geographic Information

The spatial econometrics regarding the virtual landscape activity dimension in this study aim to explore the coupling relationship between virtual spatial vitality and the physical pedestrian environment through crowdsourced data. Similarly abandoning coarse administrative village (Li) boundaries, this chapter conducts topological operations entirely within the vector data model using the "Minimum Statistical Units (MSUs)" as the foundational spatial units.

The specific spatial statistical procedure is divided into two core steps:

The first step involves spatial topological connection and point-in-polygon counting. The research team bridges the point vector layer of virtual activity

nodes (including virtual stations, big flowers, and regular player gathering nodes) extracted from the *Pikmin Bloom* augmented reality game via third-party tools with the complete polygon layer of 1,334 minimum statistical units in Xinzhuang District. Utilizing the "Count Points in Polygon" algorithm within QGIS, the total number of virtual activity nodes falling inside each micro-street block polygon grid is precisely calculated.

The second step involves functional grid normalization and choropleth mapping visualization. Because the geometric surface areas of individual minimum statistical units vary, this study invokes geometry measurement functions via the Field Calculator within the attribute table to normalize each polygon area into square kilometers. This eliminates area-induced interference and accurately reflects the digital footprint clustering intensity per unit area, generating a spatial density indicator of "virtual node count / minimum statistical unit area (km²)" for each micro-street block. Ultimately, based on the statistical densities, the research team applies choropleth mapping techniques to visualize the results using a red-gradient color scheme (ranging from white vacant zones and light-pink low-intensity areas to dark-red high-heat zones), producing the *OSM Hotspot Analysis Map of Xinzhuang District*. This output is

utilized to perform direct spatial proximity comparisons against MRT stations, major arteries, and natural river channels within the district, serving as the empirical foundation for subsequent discussions on the correlation between virtual leisure vitality and physical pedestrian safety.

5. RESULTS AND DISCUSSIONS

5.1. Spatial Differentiation and Supply-Demand Mismatch of Green Space Resources and Resident Population

According to the visualization results of this study, which integrated urban planning green space geometry and minimum statistical unit resident population data via a Geographic Information System, the spatial distribution of public green space resources and resident population density in Xinzhuang District exhibits significant heterogeneity, empirically demonstrating the phenomenon of environmental injustice in urban spatial development.

5.1.1. Spatial Geometric Characteristics of the Supply and Demand Sides

From the supply side, cross-referencing the geometric distribution of urban planning public facilities in Figure 5.1-1 reveals that the public green space resources of Xinzhuang District (encompassing parks, green spaces, children's playgrounds, plazas, and stadiums) are highly concentrated within the modern urban planning integrated development zones of North Xinzhuang, such as the Xinzhuang Sub-city Center and the Touqian Redevelopment Zone. Their spatial characteristics are manifested in intact patch areas, square geometric shapes, and exceptionally high density. Conversely, across the older urban

districts of South Xinzhuang and early-developed settlements such as Hougang, Min'an, and Xisheng, the urban planning public green spaces appear extremely fragmented and scarce, presenting large-scale blank vacuum zones of green space on the map.

Spatial Distribution of Public Green Spaces in Xinzhuang District
An Analysis Based on Urban Planning and Statistical Areas

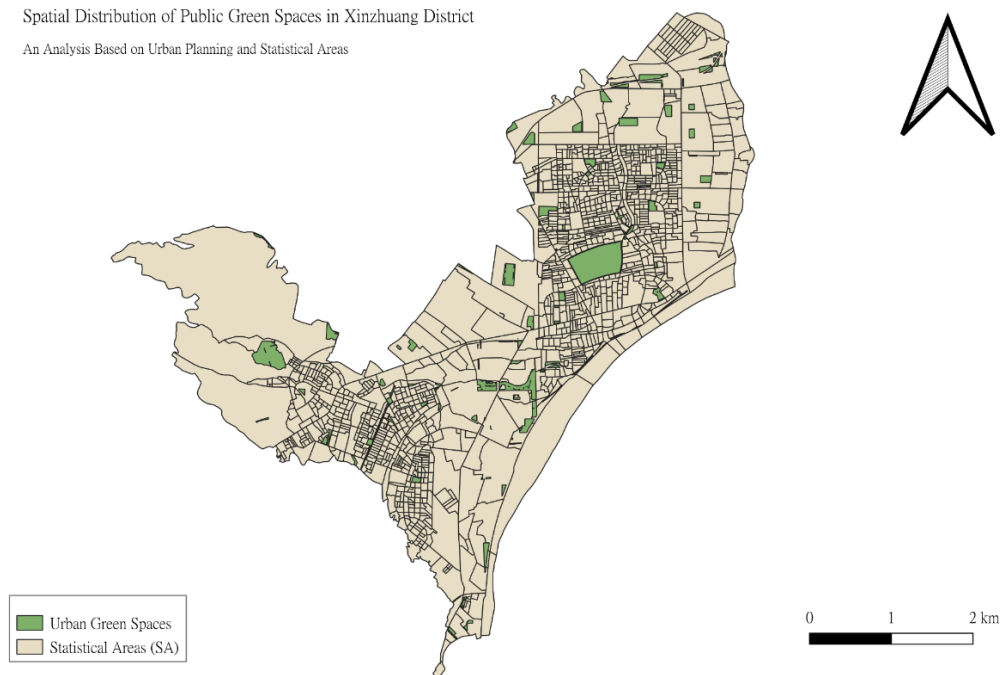


Figure 5.1-1 Spatial Distribution of Public Green Spaces in Xinzhuang District

However, cross-referencing the demand side, namely the spatial distribution choropleth map of resident population within the Minimum Statistical Units (MSUs) in Figure 5.1-2, it is clearly observable that the pattern of population clustering exhibits a reverse spatial distribution relative to green space supply. The micro-street block units of the South Xinzhuang older urban districts and traditional industrial life circles universally display a dark brown coloration, signifying high-density clustering with resident populations as high as 1,638 to 6,511 individuals per single statistical unit. This geographical reality of "high population demand, low public facility supply" empirically demonstrates that a severe spatial mismatch exists within the evolution of Xinzhuang District's

urban landscape.

Spatial Distribution of Population in Xinzhuang District

An Analysis of Demographic Density at the Basic Statistical Unit Level

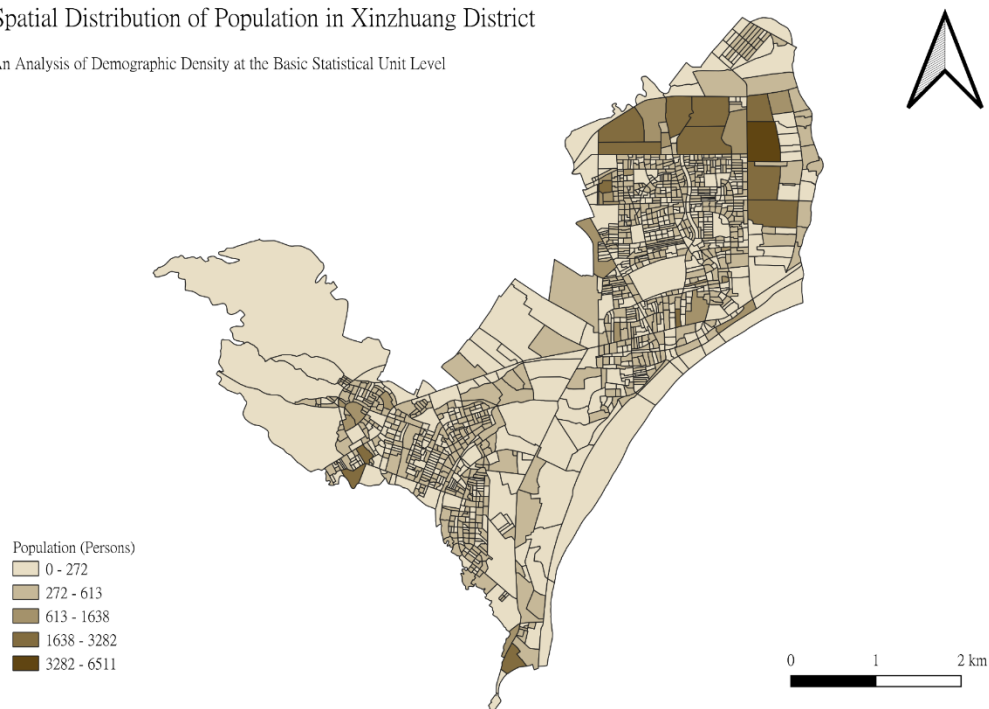


Figure 5.1-2 Spatial Distribution of Population in Xinzhuang District

5.1.2. Empirical Evidence of Environmental Injustice in Per Capita Green

Space Area

This study performed a spatial topological intersection operation on the aforementioned two heterogeneous layers, ultimately generating Figure 5.1-3, a choropleth map of the per capita park and green space area within the minimum statistical units of Xinzhuang District. This operation converted the supply-demand mismatch into a concrete environmental econometric indicator. In terms of per capita green space performance, the entire district is

fractured into polar opposite spatial landscapes between the north and the south:

First is the environmental advantage of the redevelopment zones in North Xinzhuang. Benefiting from the public facility allocation guarantees stipulated by modern urban planning frameworks enacted after 1973, large swaths of the minimum statistical units in the Sub-city Center and Touqian Redevelopment Zone are displayed in dark green, with the per capita green space area reaching or exceeding the threshold of 9.0 square meters per person. This result successfully surpasses the basic healthy urban life standard recommended by the World Health Organization (WHO), indicating that residents in this area enjoy high environmental well-being with access to the premium Zhonggang Drainage Channel landscape corridor and large sports parks.

Second is the environmental disadvantage trap of the older settlements in South Xinzhuang. Conversely, large numbers of micro-street blocks across the older urban districts of South Xinzhuang appear as a dense cluster of red (less than or equal to 2.0 square meters per person, denoting statutory green space deficiency zones) and light brown (2.0 to 4.0 square meters per person). These

values fall far below the optimal livability threshold of 4.0 square meters per person set in this study based on the average of major domestic metropolitan areas. This phenomenon profoundly reflects the historically inherited issues caused by a lack of forward-looking public facility allocation guidelines during the early period of high-density mixed residential-industrial development.

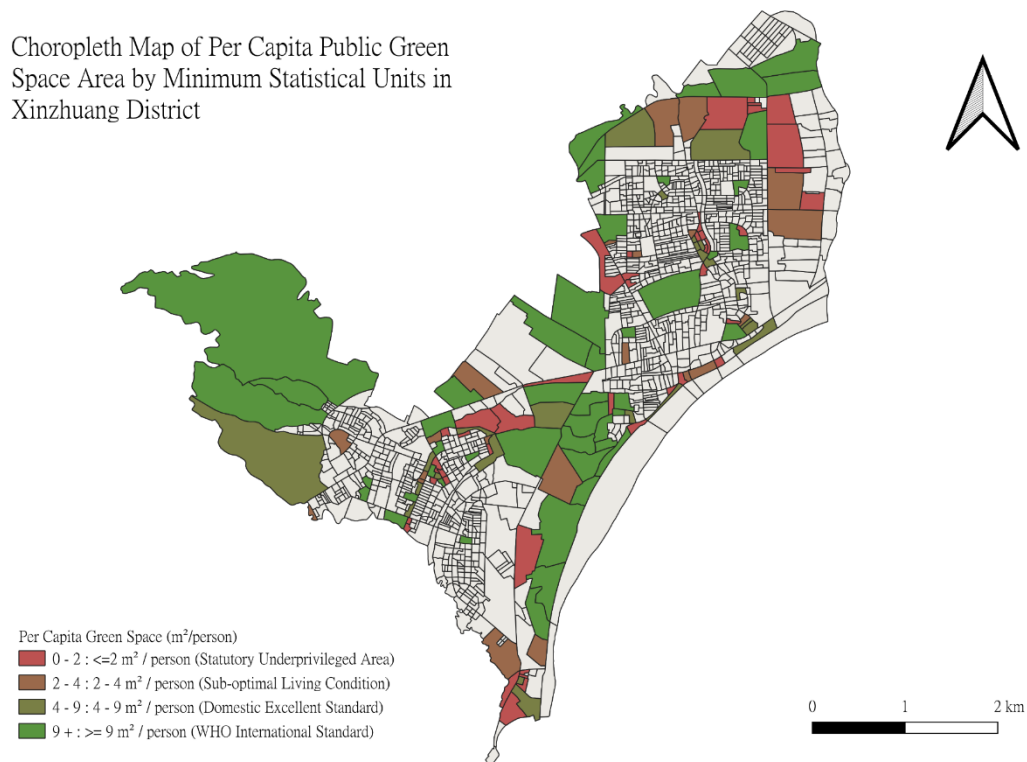


Figure 5.1-3 Choropleth Map of Per Capita Public Green Space Area by Minimum Statistical Units

5.1.3. Intertwining Conflicts of Watershed Spatial Interference and Environmental Quality

This study further examines the proximity relationship between this gap in per capita green space supply-demand and the physical watershed space. The area along the Tandigou and Taliaokeng Creek watersheds in the South Xinzhuang older urban districts not only faces a paper deficiency in per capita green space resources, appearing as a "dense red zone," but its physical residential environment is also subjected to environmental quality interference from poor water bodies. Due to the high concentration of traditional small-and-medium-sized industrial zones along Tandigou, challenges regarding poor water quality and seasonal odor issues have persisted long-term.

The 200-meter proximity interference buffer established in this study directly overlaps spatially with these residential street blocks characterized by high population density and low environmental green coverage. This explains the deep-seated conflict regarding residential livability at a micro-scale: the environmentally disadvantaged groups within the older life circles of South Xinzhuang are simultaneously beset in their daily lives by both rigid, high-heat-retention impervious surfaces and the odor interference from poorly

remediated channel sections, culminating in a typical case of environmental injustice that urgently needs to be reversed through modern urban renewal and environmental governance.

5.2 Multi-Modal Transit Integration for Low-Carbon

Transportation and Waterfront Corridor Stitching Strategies

5.2.1 Spatial Clustering Characteristics of Single Micro-Mobility and Public Transit Stations

This study first utilizes the Minimum Statistical Units (MSUs) of Xinzhuang District as microscopic spatial units to perform spatial topological joining and counting for YouBike 2.0, bus, and MRT systems. The analytical results reveal a significant clustering of transportation resources within Xinzhuang, alongside distinct east-west and north-south spatial divergence.

In terms of public bicycle supply, cross-referencing the Choropleth Map of number of YouBike 2.0 docks within the density map in Figure 5.2-1 shows that high-supply zones for bicycle docks (reaching 15 to 54 docks per single statistical unit) are heavily concentrated within the Touqian Redevelopment Zone and the Xinzhuang Sub-city Center in North Xinzhuang, as well as the

older urban street blocks along Zhongzheng Road (the MRT Zhonghe-Xinlu Line). Conversely, across the western foothills of Danfeng and Huilong on the border of Xinzhuang, as well as within certain minimum statistical units near the southern banks of the Dahan River, the supply universally falls into the low-intensity range of 1 to 4 docks, demonstrating a distinct north-south and east-west disparity in active transportation infrastructure across the administrative district.

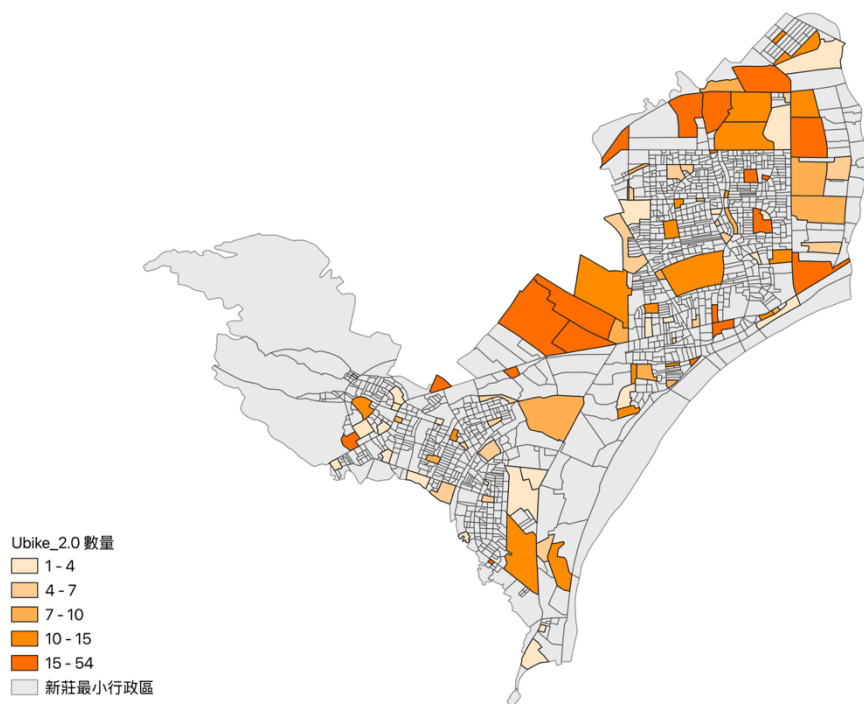


Figure 1.2-1 Choropleth Map of Number of YouBike 2.0 Docks within the Minimum Statistical Units of Xinzhuang District

Regarding bus stops and MRT stations, cross-referencing Figure 5.2-2 and Figure 5.2-3 indicates that bus stops (Figure 5.2-3) exhibit a more widespread coverage across the entire area, yet high-density zones (reaching 41 to 51 stops per single statistical unit) remain noticeably skewed toward the streets of the New Taipei Industrial Park on the northeast side and major commuter arteries. In comparison, MRT station nodes (Figure 5.2-2) are strictly constrained by rail alignment and station construction costs; the vast majority of the minimum statistical unit polygons on the map are covered in gray, which represents zero stations. This demonstrates that most micro-street blocks lack a direct rail transit presence, leaving only a few specific micro-polygon units to benefit from the direct radiation effects of MRT entrances and exits.

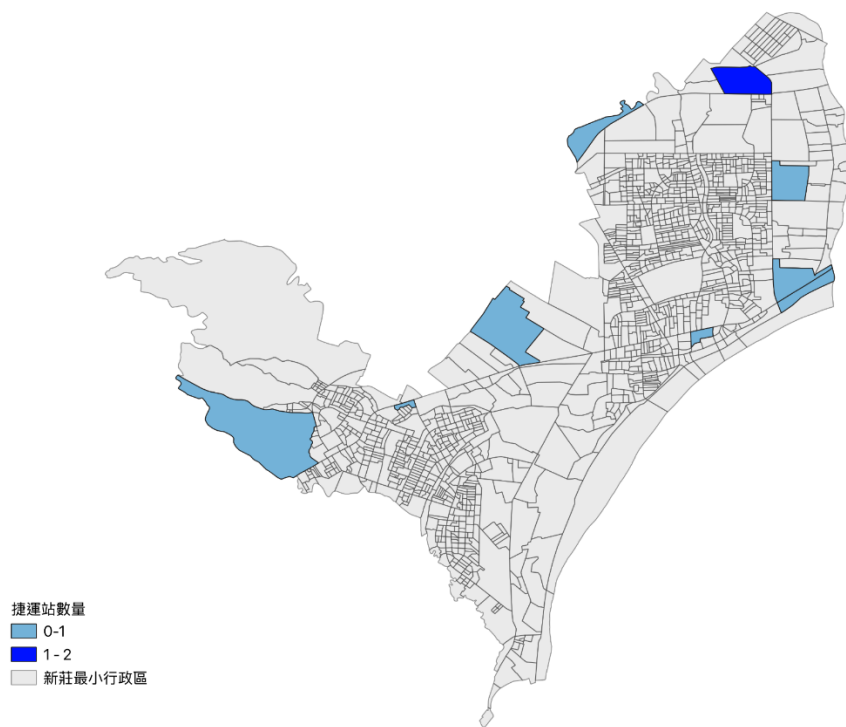


Figure 5.2-2 Choropleth Map of MRT Station Entrance/Exit Counts within the Minimum Statistical Units of Xinzhuang District

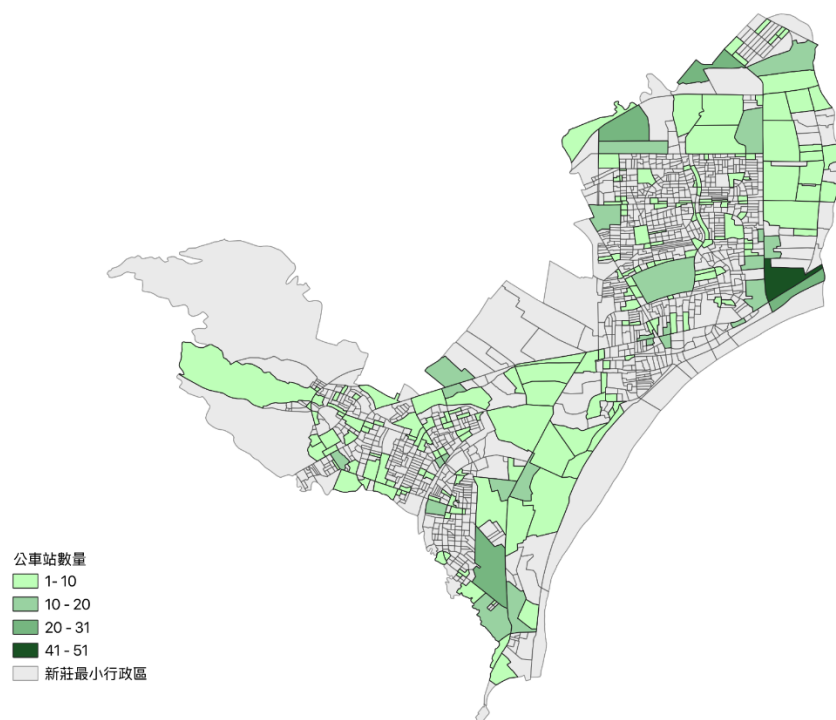


Figure 5.2-3 Choropleth Map of Bus Stop Counts within the Minimum Statistical Units of Xinzhuang District

5.2.2 Spatial Evaluation of Multi-Modal Transit Service Radius and Multi-Layered Coverage Rates

To evaluate the practical, daily convenience for urban residents in accessing different transportation services, this study establishes vector buffers based on a 600-meter theoretical service radius—a distance commonly used in public transit planning—to perform a spatial coverage assessment of multi-modal transit.

First, a review of multi-modal transit service boundaries was conducted using the active and green transportation tool distribution map in Figure 5.2-4. In this map, the 600-meter circular buffers of bus stops (green), YouBike 2.0 stations (yellow), and MRT stations (blue) are highly intertwined across the basin plains. Bus stops demonstrate global linear and surface-level coverage, while YouBike 2.0 stations exhibit dense, point-based diffusion across various residential life circles. The overlapping boundaries of these three modes clearly display the proximity and marginal expansion of different transit facilities at a microscopic scale.

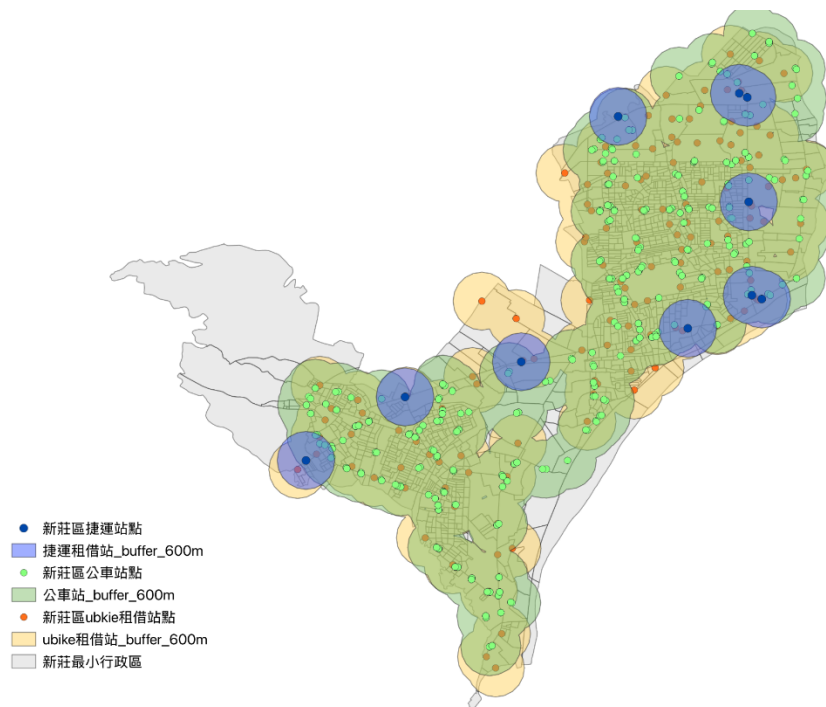


Figure 5.2-4 Vector Buffer Map of the 600-Meter Service Radius for Active, Green, and Public Transportation Modes in Xinzhuang District

To further quantify the synergistic effects of micro-street blocks simultaneously accessing multiple transit modes, this study evaluates the multi-layered networks through a two-stage vector Intersection spatial overlay operation, the results of which are shown in Figure 5.2-5:

- Bus and YouBike Dual-Composite Network (Light Purple Area):** The intersection of bus and YouBike buffers covers the vast majority of the physical space in the Xinzhuang plains (accounting for approximately 74.17% of the total area of Xinzhuang), demonstrating that these two

active transportation and highway passenger transit modes construct an exceptionally stable dual-service baseline for short-to-medium-range transfers.

- **Triple-Transit High-Efficiency Composite Transportation Core**

Network (Dark Purple Area): When further intersecting with the MRT station buffers, the dark purple area precisely delineates the core transportation hubs and high-accessibility corridors that simultaneously possess the triple-transfer functionality of MRT, bus, and YouBike.

However, this also exposes the western foothill fringes (the western foothills of Danfeng) and the southwestern boundary regions, which emerge as transit-marginal vacuum zones with lower composite public transit accessibility outside the dark purple core network. This trinity core network accounts for only 12.11% of the total area of Xinzhuang.

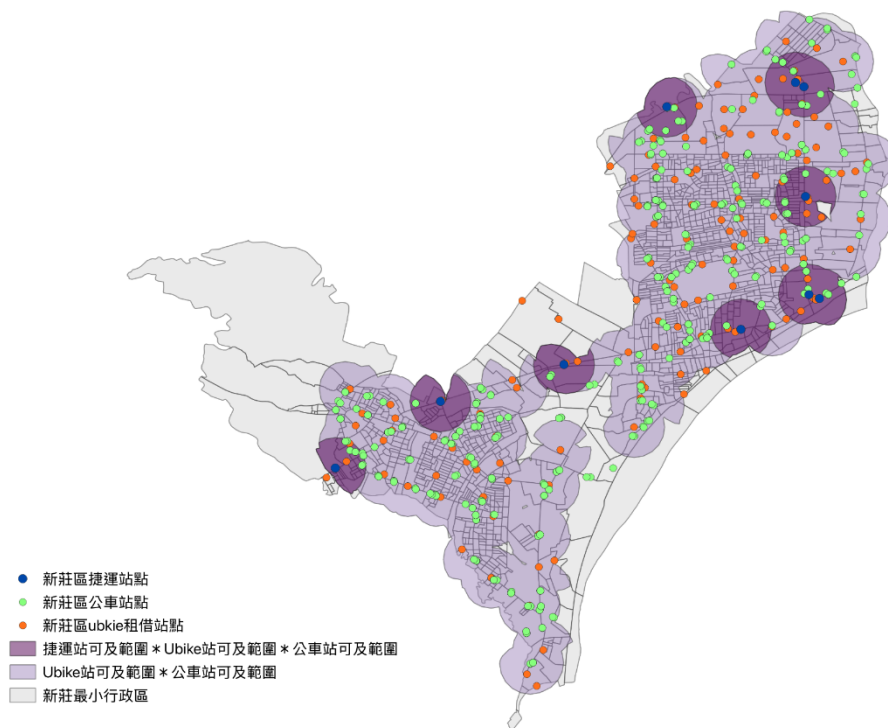


Figure 5.2-5 Multi-Stage Intersection Overlay Map of Multi-Modal Public Transit Service Scopes in Xinzhuang District

5.2.3 Cross-Border Spatial Relationships Between Active Transportation

Networks and Blue-Ribbon Waterfront Corridors

This study further overlays the internal point-based public transportation network of the city with the "physical river network" and "bicycle lane access points" to explore the substantive connectivity between the commuter road network and recreational green belts:

In terms of sectoral interactions across urban waterfront corridors, cross-referencing Figure 5.2-6 reveals that the Zhonggang Drainage Channel, which

runs longitudinally through the core of North Xinzhuan, is surrounded by an exceptionally high density of bus stops and YouBike 2.0 stations, where urban commuting and waterfront recreational spaces are highly integrated. However, along the banks of the Dahan River, situated on the southern and eastern fringes of Xinzhuan, the linear alignment of the natural river channels exhibits a more detached spatial relationship with the internal public transit network of the urban area.

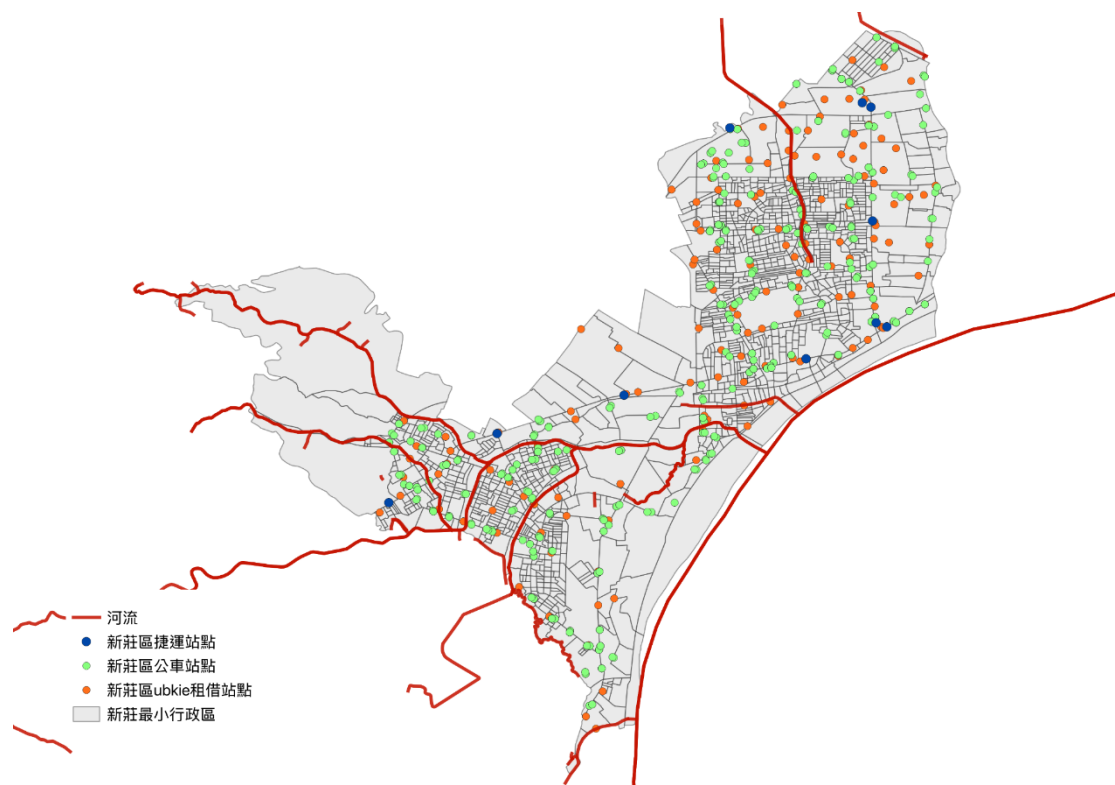


Figure 5.2-6 Overlay Map of Physical River Networks and Public Transit Station

Locations in Xinzhuang District

To dig deeper into this detachment phenomenon, the research team analyzed the service coverage of boundary-adjacent external nodes, as cross-referenced in Figure 5.2-7. It is clearly observable from the map that all bicycle lane access points (dark brown nodes) are located along the outer fringes of Xinzhuang's administrative boundary (such as the opposite bank of the Dahan River and the right-side periphery). This study generated 900-meter service buffers (brown circles) around these access points, which revealed that only a very small fraction of these external nodes' practical service radiuses brush against and cover the southern and eastern edges of the Xinzhuang administrative district. The majority of these buffer zones exhibit a significant spatial fracture with the bus stops, YouBike stations, and MRT stations inside Xinzhuang District.

This spatial relationship confirms that although Xinzhuang residents are adjacent to the Dahan River blue-ribbon corridor in terms of linear physical distance, they are constrained by high physical dike barriers and administrative boundary divisions, causing the internal commuter transit system and the external recreational active transportation system to operate in

silos. The disconnection between these boundary-adjacent external nodes and the urban street network exposes the fact that Xinzhuang residents still face high spatial thresholds if they wish to cross these barriers to enter the waterfront corridors, highlighting a lack of seamless "last-mile" transfer and guiding infrastructure.

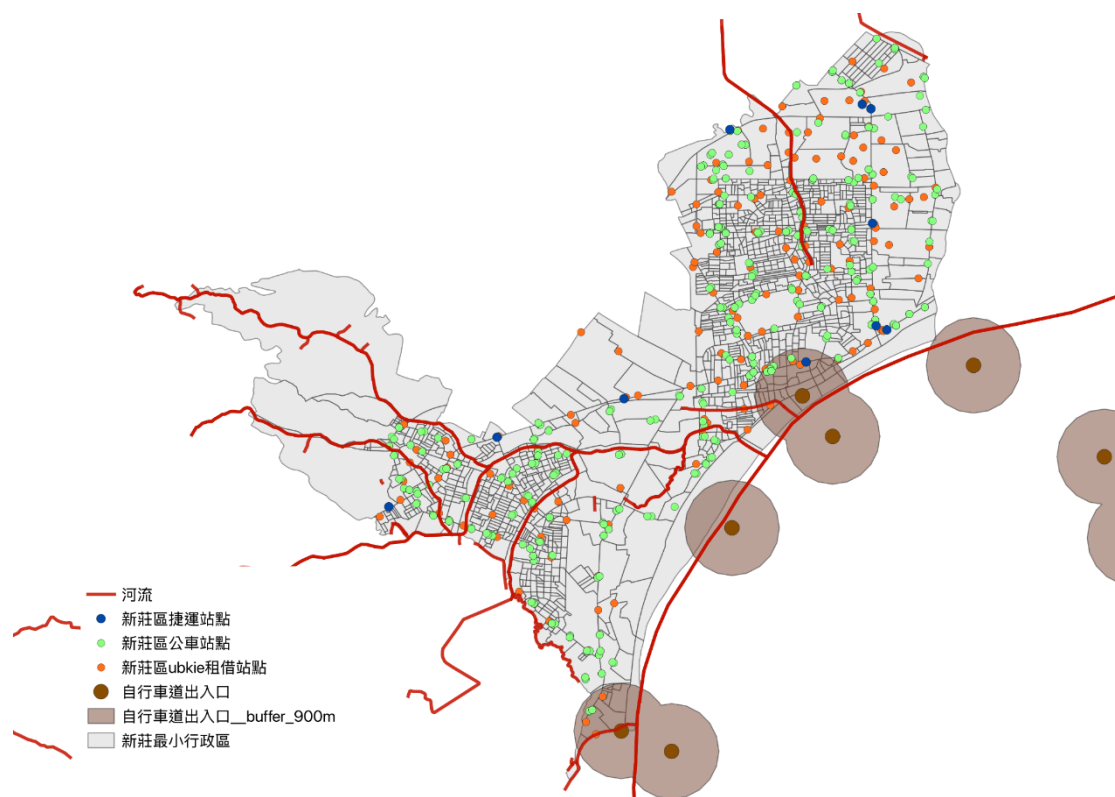


Figure 5.2-7 Spatial Relationship Between the 900-Meter Service Radius of the Riverside Bicycle Lane Access Points and the Urban Street Network

5.2.4 Practical Recommendations for Residential Location Selection Under Transportation Accessibility Constraints

High-Efficiency Transit-Oriented Premium Livable Site Selection (Dark

Purple Triple-Transit Core Area): The dark purple area, which accounts for only 12.11% of the total district area in Figure 5.2-5, represents the transport core zone possessing the highest degree of mobility freedom within the residential livability assessment. These micro-statistical units not only feature full coverage of highway passenger transit and YouBike 2.0 micro-mobility networks, but they also enjoy direct radiation benefits within the 600-meter radius of MRT stations. For car-free commuters who rely heavily on public transit and prioritize cross-district travel efficiency, this core zone offers immense residential accessibility well-being, designating it as a premium, fully saturated choice for livable housing in the model, though it also commands the highest transit-oriented development (TOD) premium in the market.

Cost-Effective, Low-Carbon Transfer Stable Life Circle Site Selection (Light

Purple Dual-Transit Coverage Area): Conversely, the light purple areas that exclude MRT stations but are widely covered by the dual intersection of bus and public bicycle networks (amounting to as high as 74.17% of the total area)

serve as "cost-effective livable zones" with exceptionally stable short-to-medium-range transfer functionalities. When potential homebuyers select sites within these zones, although they cannot access rail transit directly upon stepping out of their doors, the dense micro-mobility network constructed by YouBike 2.0 and highway passenger transit allows residents to seamlessly link to rail hubs via "last-mile" transfers with minimal time and physical effort. Housing prices in these street blocks are typically more affordable than those in the core areas, making them high-suitability locations for rigid housing demands. As for the partial transportation vacuum zones on the western and southwestern fringes that lack dual-network coverage, the site selection model classifies them as inconvenient "transportation functionality islands," where time costs must be meticulously evaluated before purchasing.

Avoiding "Pseudo-Waterfront" Defensive Site Selection for Riverside

Housing (Dahan River Boundary Fracture Zone): The disconnection of boundary nodes revealed in Figure 5.2-7 provides defensive, critical reflections for the prevailing "waterfront landscape housing" trend in the real estate market. Although residential street blocks along the Dahan River are physically adjacent to the natural blue-ribbon corridor, a distinct spatial fracture exists

because the physical dikes and external bicycle lane access points are mostly situated on the outer fringes of the district boundaries, and their 900-meter recreational buffers are disconnected from the internal micro-mobility system. Consequently, if potential homebuyers pick an edge polygon directly adjacent to the dikes under the assumption of "daily waterfront recreation," they will encounter remarkably high spatial barriers when attempting physical cross-border travel in their daily lives. This study suggests that buyers looking to integrate residential commuting with blue-ribbon recreation should prioritize micro-street blocks near the Zhonggang Drainage Channel landscape corridor or boundary locations already equipped with two-way YouBike 2.0 transfer hubs to secure substantive well-being.

5.3 Peak Convenience Within the 15-Minute City Public

Facilities Framework

To comprehensively deconstruct the spatial heterogeneity of spatial functionality across Xinzhuang District, this study adopts a dual-track vector spatial analysis model. First, vector spatial statistics are applied across Minimum Statistical Unit (MSU) coordinates to generate choropleth density maps for the five major individual dimensions alongside a consolidated,

weighted overlay (Figures 5.3-1 to 5.3-6) to precisely capture the clustering phenomena of specific facilities within micro-street blocks. Second, the discrete boundaries of micro-polygons are removed, and a pedestrian-oriented 900-meter circular service buffer is integrated. This layer is then overlaid with the OpenStreetMap (OSM) physical road base map using translucent visualization to produce the final comprehensive functional hotspot map (Figure 5.3-7).

The core empirical findings and analytical insights derived from the spatial statistics and map cross-referencing are as follows:

5.3.1 Spatial Density Heterogeneity Analysis of the Five Major Functional Dimensions

The spatial distribution of point densities across each individual dimension clearly reflects the distinct spatial preferences of public facilities and the trajectory of urban development within Xinzhuang District:

In terms of **daily shopping functionality**, cross-referencing Figure 5.3-1 reveals that high-density point zones (with the highest classification interval reaching 1,333 to 2,500 points) exhibit a highly prominent "dual-core

clustering" pattern at the minimum statistical unit level. Core One is closely tied to the historical urban axis of South Xinzhuang (the heritage street blocks centered along Xinzhuang Road and Zhongzheng Road), while Core Two manifests robustly within the Touqian Redevelopment Zone and Xinzhuang Sub-city Center to the north. This spatial phenomenon empirically demonstrates that the existing baseline population density of traditional historical settlements, coupled with the spatial influx of modern commercial chain facilities (such as PX Mart, Simple Mart, and hypermarkets) into newly emerged redevelopment zones, is dynamically driving up the daily commodity accessibility across these two major hubs.

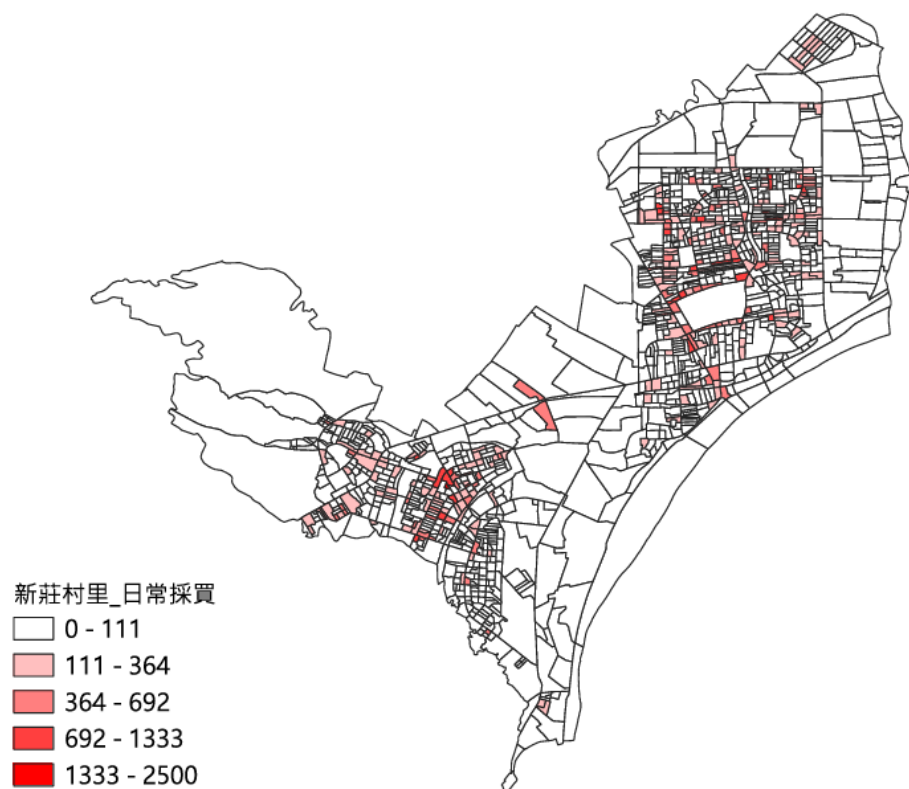


Figure 5.3-1 Choropleth Map of Daily Shopping Facility Functionality Density across Administrative Village Units in Xinzhuang District

In terms of **life support functionality** (as shown in Figure 5.3-2), which primarily encompasses the density distribution of financial institutions, post offices, and grassroots administrative agencies, the map reveals that high-density zones (333 to 1,000 points) exhibit a distinctly "linear" morphology. This pattern develops horizontally along the micro-street blocks bordering Xinzhuang's primary longitudinal transit axis (Zhongzheng Road) and forms localized, patch-like clusters in the Zhonggangcuo area surrounding the

Zhonggang Drainage Channel. This outcome demonstrates that the spatial placement of foundational urban support facilities relies heavily on the network accessibility of major traffic arteries.

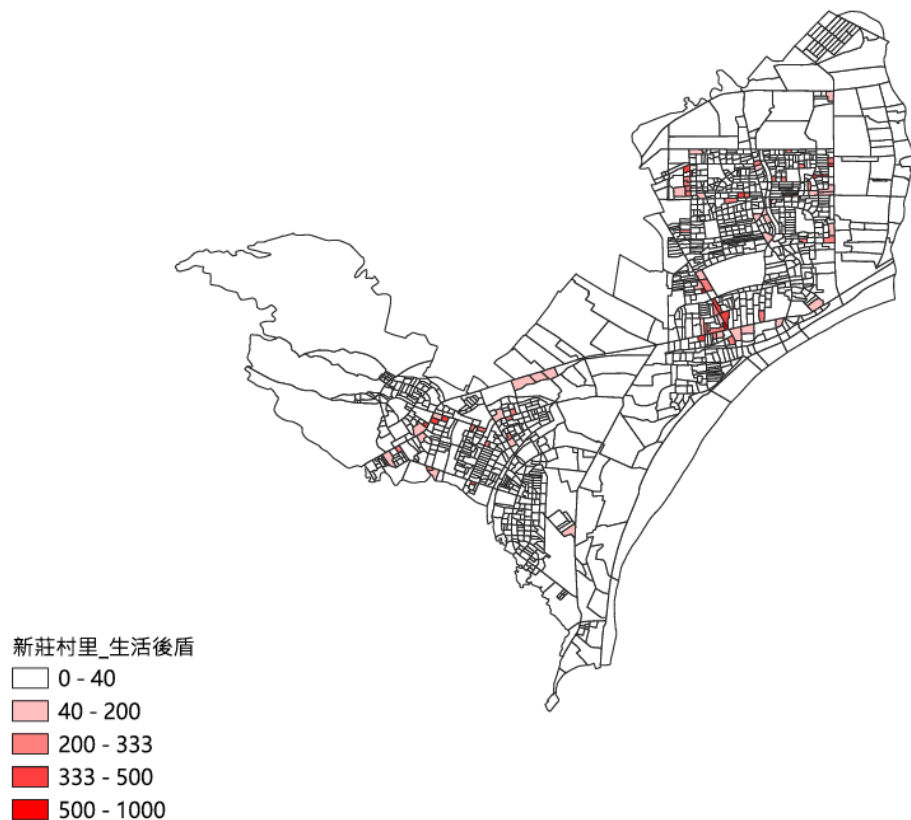


Figure 5.3-2 Choropleth Map of Life Support Facility Functionality Density across Administrative Village Units in Xinzhuang District

In terms of **childcare and cultural functionality density** (as shown in Figure 5.3-3), the spatial distribution of preschools, daycare centers, and elementary school districts (ranging from 125 to 333 points) appears relatively

decentralized. However, prominent red high-value patches emerge within the micro-polygons of Fu Jen University's Yingpan and Danfeng areas in South Xinzhuang, alongside the modern redevelopment zones and the old settlements of Zhonggangcuo in North Xinzhuang. This indicates that the allocation of educational and childcare resources is highly population-dependent, closely matching the spatial distribution of newlywed and child-rearing migrant populations, which represents a multi-point flourishing urban planning characteristic.

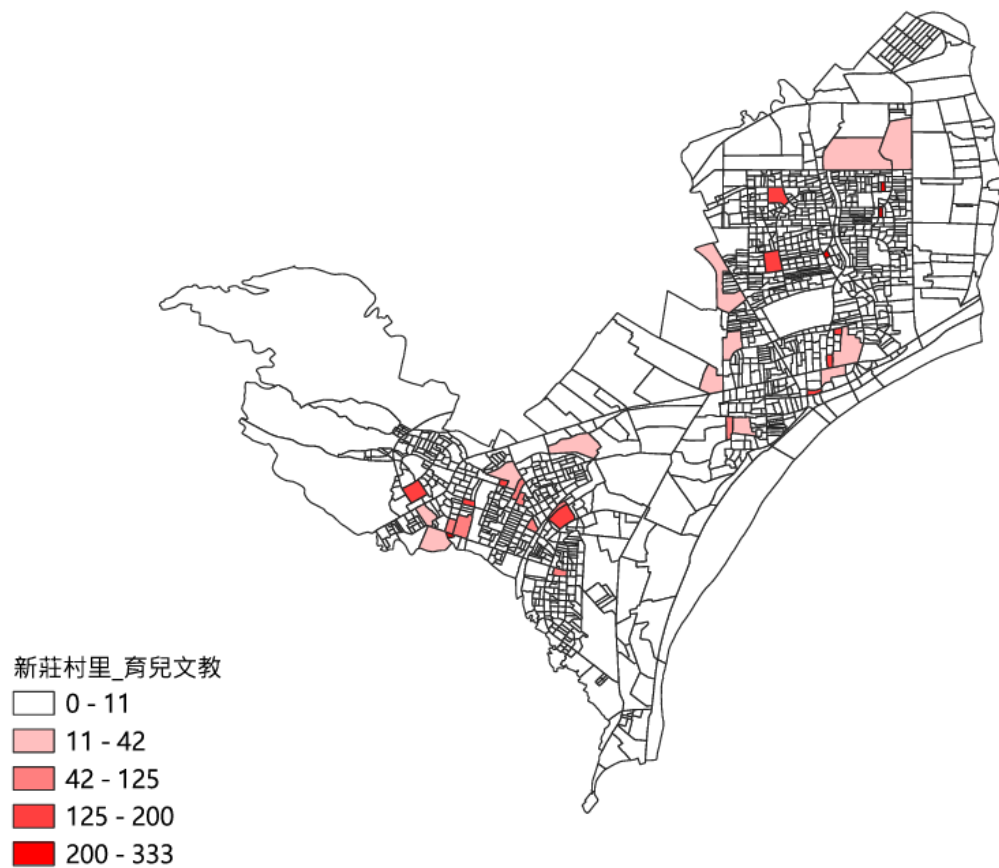


Figure 5.3-3 Choropleth Map of Childcare and Cultural Facility Functionality

Density within the Minimum Statistical Units of Xin Zhuang District

In terms of **transit commuting functionality density** (as shown in Figure 5.3-4), the public transit commuting functionality density (40 to 142.9 points) exhibits a highly pronounced polarization characteristic indicative of transit-oriented development (TOD). The high-density red street blocks radiate outward as geometric circles centered precisely on the station nodes of the MRT Xin Zhuang Line (Xin Zhuang, Fu Jen University, Danfeng, and Huilong

Stations), alongside Xingfu Station on the Circular Line and Xinzhuang Sub-city Center Station on the Airport MRT, demonstrating the absolute dominance of rail transit in shaping spatial guidance.

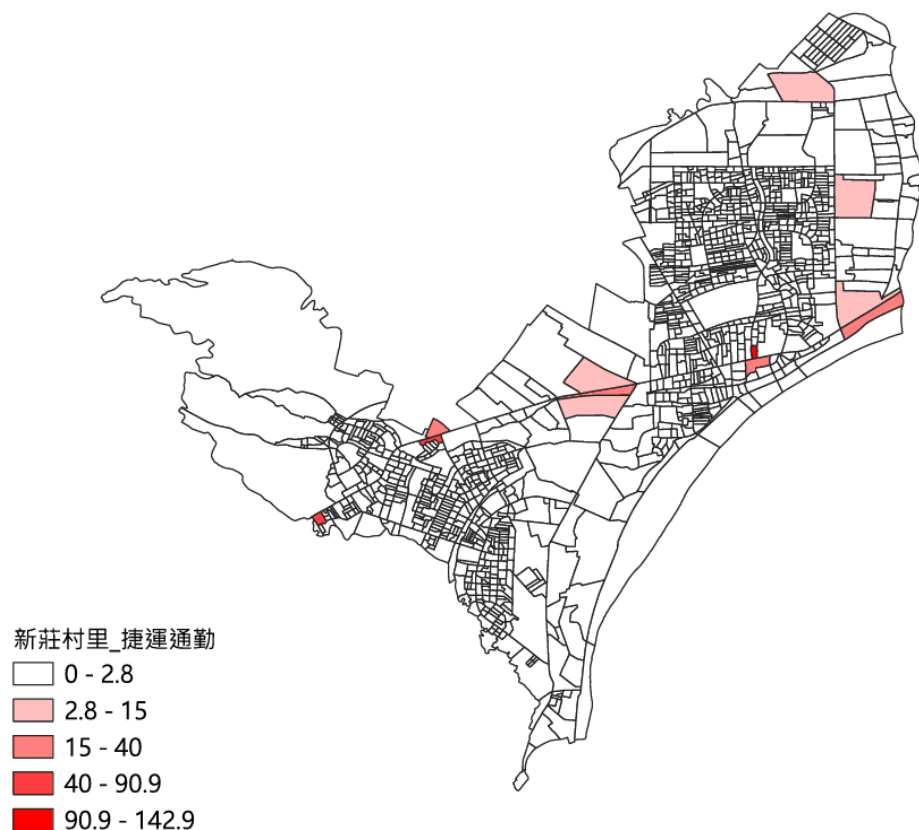


Figure 5.3-4 Choropleth Map of Transit Commuting Facility Functionality

Density within the Minimum Statistical Units of Xinzhuang District

In terms of **urban safety functionality density** (as shown in Figure 5.3-5), the distribution of high-value zones for medical, police, and firefighting infrastructure (ranging from 91 to 333 points) is the most heavily constrained. These resources are primarily concentrated around the older urban core near

the Xinzhuang MRT station and major medical centers (such as Fu Jen Catholic University Hospital and Xintai Hospital). Conversely, the remaining areas fall largely into the low-value gray interval of 0 to 14 points, strongly highlighting that the statistical units along Xinzhuang's outer periphery still require spatial allocation optimization to improve medical and disaster prevention resilience.

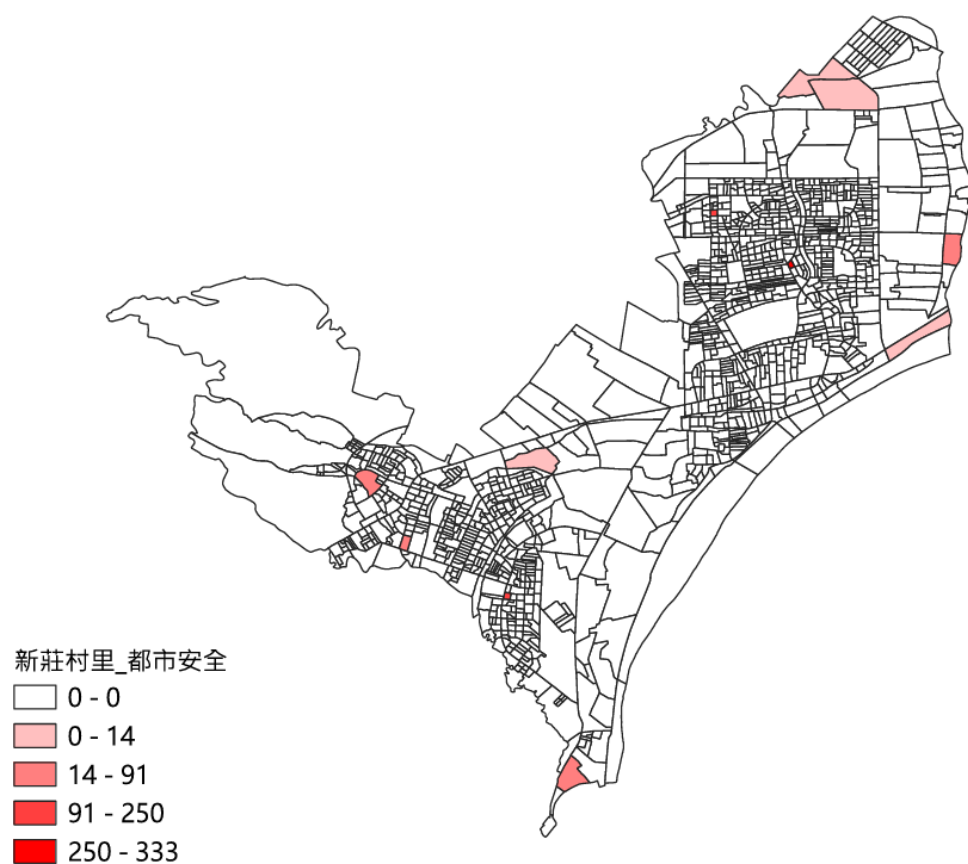


Figure 5.3-5 Choropleth Map of Urban Safety Facility Functionality Density

within the Minimum Statistical Units of Xinzhuang District

5.3.2. Spatial Heterogeneity Characteristics of Weighted Overlay for Comprehensive Facility Functionality

After performing geometric overlay and weighted summation across the five major facility layers mentioned above, we generated the weighted overlay choropleth map of comprehensive public facility functionality in Xinzhuang District shown in Figure 5.3-6. This map clearly reveals a core-periphery differentiation of comprehensive functionality throughout the entirety of Xinzhuang under the constraints of 1,334 Minimum Statistical Unit (MSU) polygons:

- **High-Score Clustering Minimum Statistical Units (1,167 to 2,500 points, deep red polygons in Figure 5.3-6):** These are heavily concentrated in the area spanning from Zhonggangcuo to Xinzhuang Station in the older urban district, alongside the Sub-city Center axis to the north, indicating that these spatial units possess highly composite urban functionalities.
- **Low-Score Peripheral Minimum Statistical Units (0 to 286 points, white and light pink polygons in Figure 5.3-6):** These are primarily distributed along the western hillside fringes of Xinzhuang (such as the outer foothills of Huilong and Danfeng) as well as within the traditional industrial zones in

the central section, reflecting a landscape where living functions are highly singular or lack the influx of public facilities.

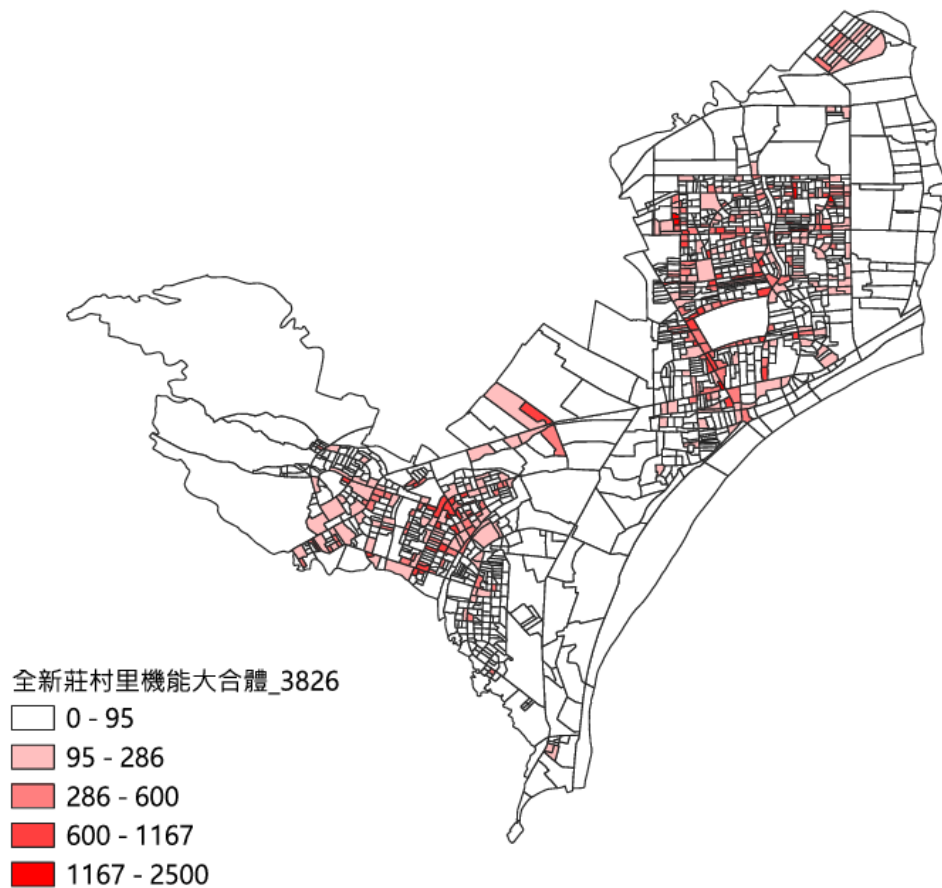


Figure 5.3-6 Choropleth Map of Weighted Overlay Density for Comprehensive Public Facility Functionality in Xin Zhuang District

5.3.3 Empirical Spatial Evidence of Comprehensive Functional Service

Buffer Hotspots Within the 15-Minute City Framework

By removing the discrete, rigid boundaries of the minimum statistical unit polygons and performing vector overlay and masking with the physical geographic base map using a 15-minute pedestrian threshold (a spatial service buffer with a 900-meter radius), we generated the 15-Minute City Comprehensive Functional Service Buffer Hotspot Map for Xinzhuang District in Figure 5.3-7. Through this empirical output, this study precisely identifies the "Peak Convenience Points (Golden Functional Cores)" within Xinzhuang District:

- **North Xinzhuang—Dual Golden Functional Peaks (Deep Red Zone in the Upper Half of Figure 5.3-7):** This zone precisely covers two major areas: the "Xinzhuang Sub-city Center" and "Zhonggangcuo." As clearly visible through the base map along major transit arteries such as Zhongzheng Road, Zhonghua Road, and Siyuan Road, this dual-peak region benefits from modern urban planning, high-density PX Mart/hypermarket commercial clusters, comprehensive public childcare and cultural facilities, and the intersection of the MRT Circular Line and the

Airport MRT. Within a 15-minute walk (or a 3-minute YouBike micro-mobility trip), residents can seamlessly access all core facilities, yielding comprehensive convenience scores that rank highest across the entire district.

- **South Xinzhuang—Linear Functional Peak (Deep Red Zone in the Lower Half of Figure 5.3-7):** Another prominent convenience peak extends westward along the axis of Zhongzheng Road, aligning precisely with "Yingpan (the Fu Jen University Commercial District)" and the "Huilong and Danfeng Station Cores." This area exhibits powerful transit-corridor-driven characteristics, achieving near-perfect scores for public transportation and daily shopping functionality, establishing itself as the most stable livable core in South Xinzhuang.
- **Functional Peripheries and Policy Optimization Zones (Light Pink to Outer Boundaries in Figure 5.3-7):** The areas where coloration noticeably fades on the map are primarily distributed along the hillside fringes bordering Taishan and Linkou, the western foothills of Huilong, and the industrial zones on both the northern and southern flanks. Although these sectors maintain basic shopping or bus coverage along main corridors, they exhibit clear deficiencies within the 900-meter buffer coverage for life

support (financial and administrative services) and urban safety (large-scale medical facilities), designating them as "functional vulnerability zones" that future urban planning frameworks must prioritize for infrastructure reinforcement.

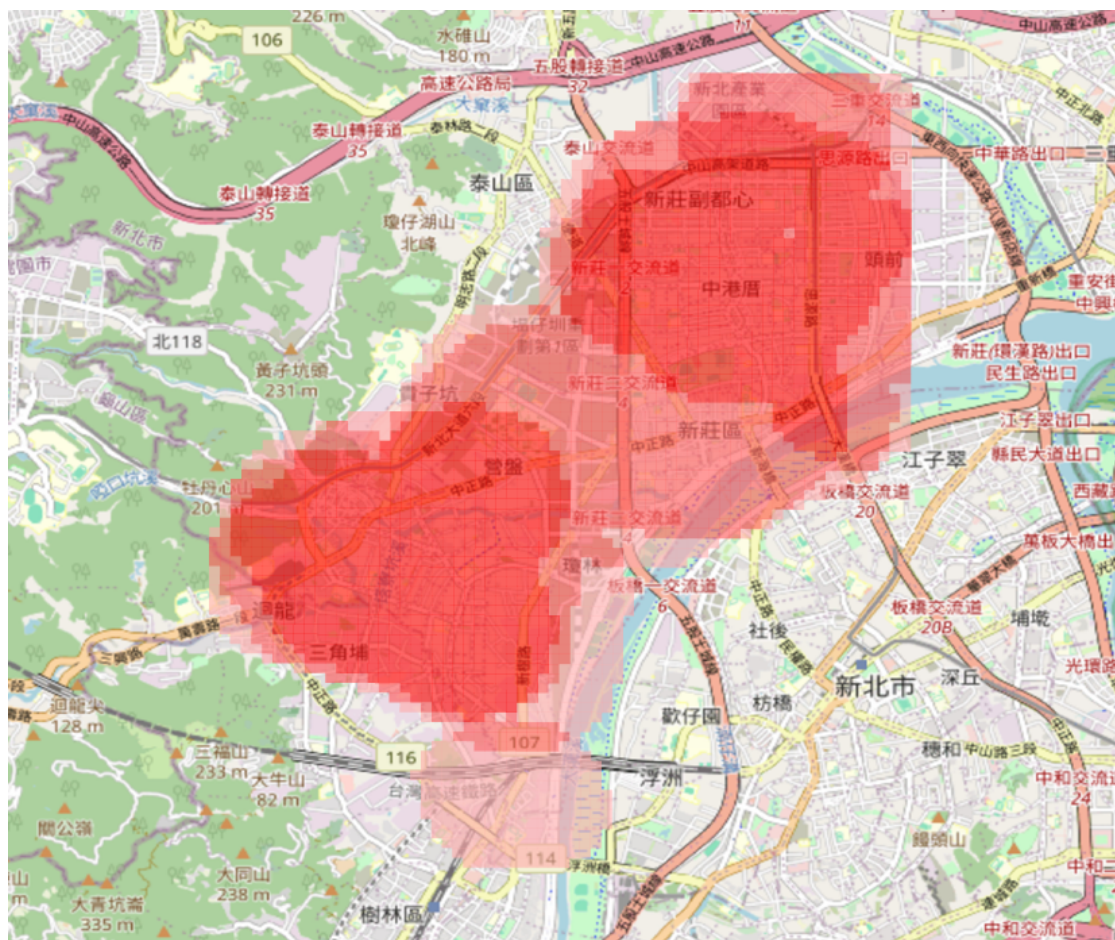


Figure 5.3-7 Comprehensive Functional Service Buffer Hotspot Map within the 15-Minute City Framework of Xinzhuan District

5.3.4 Practical Evaluation of the 15-Minute City Urban Life Functionality for Residential Site Selection

The core academic and practical value of the visualized spatial evidence generated by this study through multi-layer geographic information system (GIS) intersection overlays (as shown in Figure 5.3-7) lies in its ability to directly provide a data-driven geographic support system for potential homebuyers, housing development policies, and micro-scale residential site-selection decisions across the entire Xinzhuang area:

Residential Environmental Premium Analysis of High-Convenience Golden

Functional Peaks: The areas exhibiting saturated deep-red zones in Figure 5.3-7—namely the Sub-city Center, Zhonggangcuo, and the Yingpan (Fu Jen University) life circles in South Xinzhuang—represent top-tier, functionally impeccable site selections in urban livability evaluations. These spatial units highly integrate public transit (TOD benefits) with daily shopping resources. For homebuyers who prioritize micro-mobility efficiency and the accessibility of childcare and cultural facilities (such as young families or elderly retirees), these functional peaks provide substantial environmental well-being. However, planners and potential homebuyers must also consider that such high-

livability zones often command a significant environmental house price premium in the market, signifying a higher barrier to entry alongside high resource concentration.

Residential Suitability Reinforcement and Transformation Potential of

Functionally Vulnerable Peripheral Zones: Conversely, the areas where the coloration noticeably fades and service buffer coverage is insufficient—such as the hillside fringes bordering Taishan and Linkou, the western foothills of Huilong, and the northern and southern industrial peripheral street blocks—are classified as functionally vulnerable zones in residential livability evaluations. These sectors face substantive spatial gaps in access to administrative life support and large-scale medical safety infrastructure.

Nonetheless, from the perspective of urban renewal and social housing site selection, these low-scoring peripheral zones offer relatively affordable land costs or housing prices, making them suitable for non-profit residential spaces or budget-constrained buyers. Looking ahead, housing policies should move away from indiscriminate macro-level construction and instead champion "urban acupuncture"—the precise insertion of micro-facilities (such as introducing community daycare spots, multi-functional pharmacies, and

micro-scale public bicycle transfer stations directly into low-scoring blocks).

This approach can substantially mend spatial inequalities with minimal public investment, fulfilling the last mile of residential livability for Xinzhuang's peripheral neighborhoods.

5.4 Spatial Analysis of Traffic Accident Risks Within the Road Network

This study's environmental risk assessment regarding the road traffic safety dimension focuses on identifying negative constraint factors within urban spaces. By applying Jenks Natural Breaks optimization to the density of Class A2 traffic accidents along the road network for the year 2025 (Year 114 of the Republic), the Spatial Classification Map of Traffic Accident Risks for the Road Network in Xinzhuang District, New Taipei City, was generated as shown in Figure 5.4-1. The analytical results demonstrate that Xinzhuang's traffic safety landscape exhibits highly pronounced characteristics of "arterial polarization" and "residential-industrial mixed-use conflicts."

Traffic Accident Risk Network Analysis Map of Xinzhuang District, New Taipei City

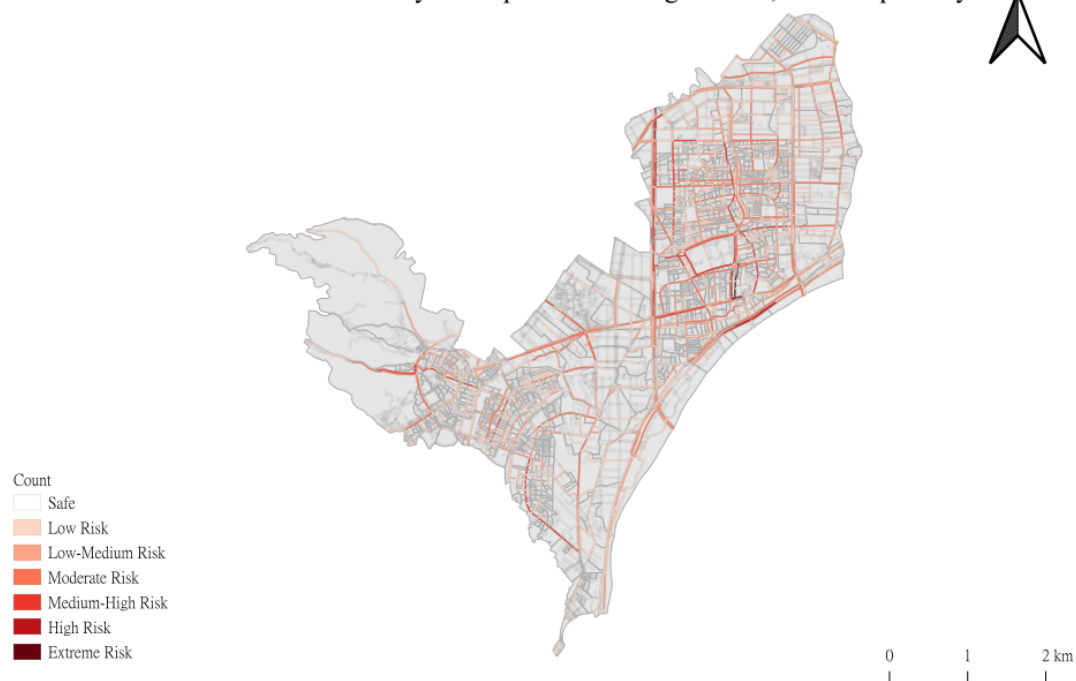


Figure 5.4-1 Traffic Accident Risk Network Analysis of Xinzhuang District, New Taipei City

5.4.1 1 Empirical Evidence of Composite Spatial Hazard Hotspots

Based on the high-risk spatial clustering patterns shown in Figure 5.4-1, the traffic hazard hotspots across the entirety of Xinzhuang are primarily driven by two major urban landscape dynamics:

- **High-Conflict Street Blocks with Mixed Industrial and Residential**

Traffic: Due to their close proximity to industrial zones, the corridors surrounding Siyuan Road and Huacheng Road experience severe traffic mixing among large heavy-duty trucks, logistics vehicles, and peak-hour

commuting motorcycles. Under road network analysis, all primary intersection nodes along these sections fall into the Extreme Risk category, confirming that "mixed vehicle types" and "excessive through-traffic" are the primary causes of frequent accidents on the eastern side of Xinzhuang.

- **The "Linear Guidance Effect" of Major Commuting Arteries:** The core transportation arteries cutting through Xinzhuang—Zhongzheng Road (the axis of the MRT Xinzhuang Line) and New Taipei Boulevard to the north—serve as mega-commuting routes connecting Taipei City, Sanchong, and the Taoyuan area, handling massive traffic volumes around the clock. The spatial distribution on the map clearly demonstrates that any minimum statistical units whose geometric boundaries are immediately adjacent to these primary commuting arteries experience a significant spike in accident density. This creates highly conspicuous, deep-red linear bands across the space that slightly radiate into the neighboring street blocks.

5.4.2 Geometric Barrier Analysis of Spatial Safe Zones

Conversely, the spatial units in Figure 5.4-1 that predominantly feature "Safe" and "Low Risk" classifications demonstrate the positive protective benefits of urban planning and topographic barriers:

- **Barrier Protection via Natural Topography and Low Road Density:** The large "Safe" zone stretching along Xinzhuang's western border is primarily composed of the areas tucked against the foothills of Danfeng and Huilong. Due to the topographic constraints of hillside terrain, this belt features lower urban development density, a sparse physical road network, and sits outside the primary corridors traversed by commuting populations, resulting in an exceptionally safe, low-risk state in spatial econometrics.
- **Benefits of Pedestrian-Vehicle Segregation and Gridiron Road Networks in Modern Redevelopment Zones:** Certain newly emerged redevelopment zones located in North Xinzhuang (such as the interior residential street blocks of Touqian and the Sub-city Center), alongside select purely residential minimum statistical units far removed from arterial roads, benefit from the gridiron street layouts of modern urban planning. These configurations offer excellent sightlines at intersection geometries and incorporate comprehensive pedestrian setbacks. Consequently, the traffic risk rating in these areas is effectively suppressed to the "Low Risk"

interval, empirically validating the substantial synergistic effect of high-quality urban design in mitigating traffic risks.

5.4.3 Practical Residential Site Selection Decisions Under Road Traffic Risk

Constraints

Synthesizing the spatial distribution of traffic accident risks from Figure 5.4-1 provides potential homebuyers and site-selection practitioners with the following three concrete, practical livability decision support strategies:

Pedestrian-Friendly Safe Greenbelt Site Selection Strategy (Safe to Low

Risk Zones): The "Safe" site selection spaces in Figure 5.4-1 are mainly concentrated in the foothill street blocks of Danfeng and Huilong in Lower Xinzhuang, as well as the interior pocket neighborhoods of purely residential communities far removed from major longitudinal traffic arteries. These micro-units perfectly bypass the blind spots of high-volume commuting dynamics, offering a slower pace of life, high pedestrian friendliness, and exceptionally low routine traffic noise and vibration interference. In multi-criteria site-selection models, such street blocks represent premium residential choices equipped with immense environmental well-being and personal safety.

Redevelopment Zone "Interior-Line Site Selection" Strategy Balancing

Functionality and Safety (Medium to High Risk Zones): For newly developed street blocks distributed across the Xinzhuang Sub-city Center and Touqian Redevelopment Zone that appear yellow (Medium Risk) to orange (High Risk) on the map, these locations simultaneously possess a high density of public facility convenience (the golden peaks discussed in Section 5.3). This study recommends a spatial risk-hedging strategy of "purchasing commercial offices along major avenues, but choosing homes along interior lines." When selecting a property, buyers should precisely target residential units situated on secondary roads or within closed pocket blocks inside the redevelopment zones. This move allows residents to enjoy the dual convenience of walking to MRT TOD stations and modern commercial districts while utilizing physical geometric distance to effectively steer clear of the "Extreme Risk" accident hotspots at major arterial intersections.

Double-Track Assessment for Saturated-Functionality Extreme Risk Zones:

The MRT-adjacent street blocks along Zhongzheng Road, along with the mixed residential-industrial zones on Huacheng Road and Siyuan Road to the east, form highly concentrated, deep-red "Extreme Risk" segments in Figure

5.4-1. Due to early development, these sectors possess high daily shopping convenience and a dense concentration of comprehensive facilities. If homebuyers choose to invest here due to resource dependency, this study recommends a double-track site assessment utilizing both "internal hardware defense" and "external engineering transitions":

- **Internal Hardware Defense (Noise Mitigation and Insulation):** Spatial site selection should prioritize units on middle-to-high floors. The building facade must be equipped with standard high-insulation airtight windows and energy recovery ventilators (ERVs) to use physical structures to block the round-the-clock traffic honking, noise, and residential well-being disruptions stemming from traffic accidents along major arteries.
- **External Engineering Micro-Site Selection (Risk Avoidance):** The benefits of intersection geometric improvements should be incorporated into the external evaluation weight. Taking the pedestrian-oriented traffic improvement project at the intersection of Xingfu Road and Siyuan Road completed in September 2025 as an example—which included reducing intersection corner radiuses, adding pedestrian refuge islands, simplifying dedicated motorcycle left-turn signal phases, and extending sidewalks—

nodes that have undergone such pedestrian-centric retrofits have seen their actual collision risks enter a downward trajectory. Homebuyers should prioritize these MRT-adjacent street blocks that carry engineering improvement dividends as a key metric for evading systemic traffic risks.

5.5 Figure 5.5 (Supplementary Indicator): Associative Characteristics Between Volunteered Geographic Information and Physical Human-Centric Pedestrian Safety

***Note: This part is not included in the final decision-making process.*

This study innovatively introduces crowdsourced Volunteered Geographic Information (VGI), utilizing the discrete locations from the augmented reality (AR) game *Pikmin Bloom* as digital footprint indicators. Point counting and normalized spatial density estimations were conducted across Minimum Statistical Unit (MSU) coordinates using QGIS. By cross-referencing and overlaying Figure 5.5-1 (Hotspot Analysis Map of Volunteered Geographic Information within the Minimum Statistical Units of Xinzhuang District) with Figure 5.5-2 (Physical Geographic Cross-Reference Base Map of Xinzhuang District), this section aims to deeply explore the topological coupling

relationships among virtual spatial vitality, physical commercial landscapes,
and human-centric environmental safety.

Spatial Density Analysis of Pikmin Bloom Hotspots,
in Xinzhuang District

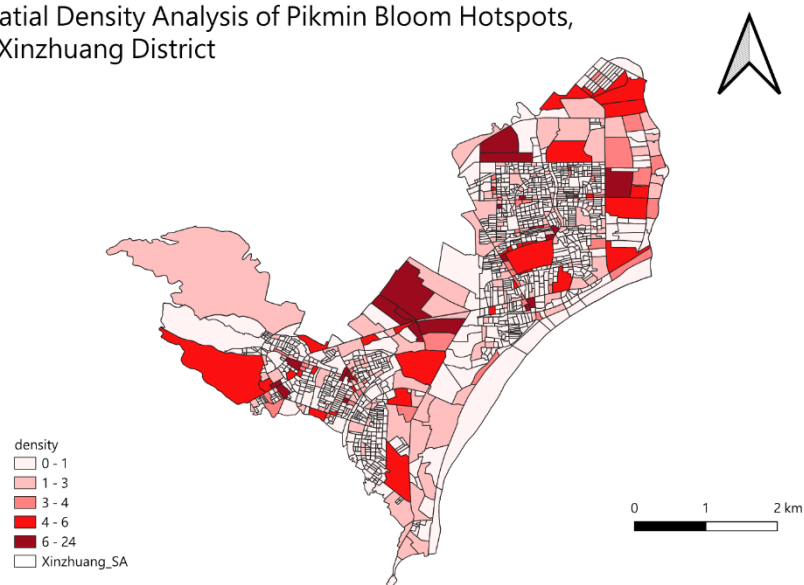


Figure 5.5-1 Hotspot Analysis Map of Volunteered Geographic Information
within the Minimum Statistical Units of Xinzhuang District



Figure 5.5-2 Physical Geographic Cross-Reference Base Map of Xinzhuang

District (Cross-Referenced with Google Maps Environmental Cartographic Data)

5.5.1 Spatial Clustering and Geographical Causal Analysis of Core

Observation Points

Based on the geometric alignment between Figure 5.5-1 and Figure 5.5-2, the digital footprint hotspots within Xinzhuang District exhibit a highly pronounced spatial heterogeneity, which can be explicitly categorized into two extreme landscape types:

- **High-Density Commercial Coupling Cores Around Transit Hubs (High-Hotspot Zones):** Cross-referencing the right half of Figure 5.5-1 clearly reveals that the Minimum Statistical Unit (MSU) coordinates surrounding the MRT Zhonghe-Xinlu Line (Touqianzhuang and Xinzhuang Stations), the Circular Line (Xingfu Station), and the New Taipei Industrial Park Station collectively exhibit the most saturated deep-red coloration (with density intervals reaching 6 to 24 points/km²). This result empirically demonstrates a high positive correlation between virtual activity hotspots and core public transit nodes (MRT Nodes). Serving as urban "passenger magnets," these stations not only guide the rigid commuting population, but their

surrounding high-density living facilities (such as convenience stores, clinics, cafes, and restaurants) reflected by OpenStreetMap (OSM) data also generate a powerful spatial coupling with the in-game nodes. This phenomenon deeply confirms that the layout of the virtual landscape relies heavily on spaces in the physical world that simultaneously possess "high pedestrian mobility" and "diverse, composite commercial typologies." In essence, the clustering of virtual vitality remains a spatial reflection of the physical micro-economy.

- **Boundary Effects and Functional Buffer Zones Along the Dahan River**

(Low-Hotspot Zones): Conversely, along the fringes of the Dahan River on the southern and eastern edges of Xinzhuang (the vicinity of Huanhan Road and the river dike roads), the minimum statistical units appear as a vacuum of white and light pink in Figure 5.5-1 (falling within the extremely low density interval of 0 to 1 point/km²). In urban geography, this represents a classic manifestation of the "edge effect" and "functional isolation." Physically serving as a high-speed vehicular artery, Huanhan Road lacks pedestrian-friendly spaces and micro-living functions (containing very few OSM nodes). Furthermore, the towering concrete dikes spatially block any pedestrian extensions by urban residents.

Consequently, a "high traffic flow, zero pedestrian flow, zero dwelling time" urban vitality vacuum has formed along this corridor, exhibiting a dual-low phenomenon across both physical human-centric and virtual indicators.

5.5.2 Developmental Pattern Comparison of Xinzhuang MRT and Population Evolution (TOD vs. Demand-Driven)

Further exploring the spatial causes behind the deep-red, high-score minimum statistical units in Figure 5.5-1 reveals that Xinzhuang District incorporates two distinct urban landscape evolution models, using "new vs. old" geographical sub-districts as a prominent watershed:

- **Traditional Population Demand-Driven Model (Population Accumulation Preceding Rail Transit):** This model is primarily reflected in the street blocks surrounding Xinzhuang Old Street (Xinzhuang Station) and the Fu Jen University Commercial District (Fu Jen University and Danfeng Stations). Its developmental logic follows a trajectory of "early population concentration leading to transit congestion, followed by MRT construction for relief." As the historical birthplace of Xinzhuang, this region had accumulated a dense residential population and spontaneous

older commercial districts decades before the MRT opened. Within these minimum statistical units, Pikmin hotspots overlap highly with historical sites and traditional shophouses, demonstrating the population depth and routine pedestrian flow brought about by historical landscapes.

- **Transit-Oriented Development Model (TOD-Oriented: Rail Transit**

Guiding Subsequent Population Influx): This model precisely corresponds to the newly emerged street blocks around Xinzhuang Sub-city Center, Xingfu, and New Taipei Industrial Park Stations. Its developmental logic operates on "planning the transit network first, guiding urban development second, and attracting population influx last." Previously consisting mostly of agricultural land or traditional sheet-metal factories, the government utilized urban planning to prioritize transportation networks while introducing the Central Government Joint Office Building and large-scale commercial facilities (such as Honhui Plaza), successfully guiding high-density private real estate development. Within these minimum statistical units, modern gridiron sidewalks and pedestrian setbacks provide an excellent physical carrier for digital footprints, causing the virtual hotspots to manifest as clean, square, block-like clusters.

5.5.3 Practical Residential Site Selection Decisions Under Virtual Hotspot

Indicator Constraints

This study considers the density of Pikmin nodes in Figure 5.5-1 as an inverse empirical indicator for measuring the "friendliness of micro-scale street pedestrian environments." For potential homebuyers and property site-selection practitioners, this digital footprint hotspot map offers two substantial interpretations for livability site selection:

- **Primary Site Selection for Ultimate Human-Centric Pedestrian Zones**

(Deep Red High-Density MSU Street Blocks): The deep-red street blocks in Figure 5.5-1 essentially represent environments with the highest safety, most developed pedestrian infrastructure, and richest abundance of "Street Eyes" across the district. Residents walking within these blocks can perfectly access high-density physical living functions within a 15-minute window. Moreover, due to comprehensive pedestrian-vehicular segregation or well-designed arcades (Qilou), pedestrian safety and well-being are exceptionally high. For homebuyers prioritizing "daily car-free pedestrian livability" (such as families with school-aged children or elderly relatives who need routine strolls), these high-density cluster zones serve

as the most flawless positive selections within a multi-criteria evaluation (MCE) site-selection model.

- **Defensive Site Selection to Avoid High-Thermal Vehicular Isolation**

Zones (White and Light Pink Low-Density MSU Street Blocks):

Conversely, if a prospective property is located along Huanhan Road on the periphery of the Dahan River or within large, traditional industrial zones inside the low-density white blocks, buyers must adopt a defensive cognitive approach. The vacuum of digital footprints within these blocks directly reflects severe physical realities: a lack of sidewalks, intensive heat accumulation due to rigid impermeable pavements, and frequent mixed heavy truck traffic (as detailed in the Section 5.4 traffic accident hotspots). Although these developments may heavily market river views or price advantages, their micro-residential environments suffer a substantive downgrade in human-centric walking, street safety, and daily recreational well-being, designating them as livability-weakened zones that warrant point deductions in MCE residential site-selection decisions.

6. CONCLUSION

6.1 Optimal Residential Site Selection: The "Golden Balance Zone" Along the Midstream of the Zhonggang Drainage Channel

Within the Multi-Criteria Evaluation (MCE) spatial overlay model across all research groups, the optimal livable zone—achieving the highest cumulative scores and demonstrating the most coherent spatial clustering—is precisely located along the residential belts flanking the midstream of the **Zhonggang Drainage Channel** (primarily encompassing the Xingfu Road Commercial District down to the street blocks bordering the western edge of the Touqian Redevelopment Zone). This region exhibits three irreplaceable balancing advantages in terms of comprehensive spatial well-being:

- **Perfect Scores in Facility Convenience and Micro-Mobility Accessibility:**

This linear street block is fully integrated into the dual golden peaks of the 15-minute city comprehensive functionality framework empirically verified in Section 5.3. Furthermore, it possesses the district's highest density of YouBike 2.0 docking pillars and dual service buffer coverage from highway

passenger transit, yielding exceptional efficiency in daily commodity acquisition and transit transfers.

- **Premium Waterfront Benefits Shielded from Basin Fluvial Disruption:**

This area highly captures the blue-green corridor landscape benefits following the remediation of the Zhonggang Drainage Channel. Compared to South Xinzhuang's older urban core—which suffers from severe proximity-based odor nuisances from traditional industrial activities and the incomplete remediation of the Tandigou channel segments (the 200-meter environmental conflict buffer zone detailed in Section 5.1.3)—the midstream residential belt of the Zhonggang Drainage Channel provides excellent micro-climate regulation and premium public space quality.

- **Flawless Evasion of High-Risk Blind Spots Along Major Transit Arteries:**

Crucially, this zone is deeply nestled within the interior of the redevelopment zone or secondary arterial blocks. Its spatial geometry successfully steers clear of the "Extreme Risk" (deep red) traffic accident hazard hotspots tracking along primary commuting arteries like Zhongzheng Road and New Taipei Boulevard. Consequently, residents can thoroughly enjoy rich urban functionalities while securing substantive

protection for personal pedestrian safety and routine psychological well-being.

6.2 Sub-Optimal Livable Site Selections and Potential Environmental Risk Reminders

This study's multi-criteria model also delivers spatial decision support anchored in defensive site-selection thinking for Xinzhuang's other two primary real estate development sub-districts:

South Xinzhuang Older Urban Core (Interwoven Fabric of Saturated Functionality and Environmental Risks): This sub-district scores near-perfect marks in public facility convenience and population baseline density within historical commercial clusters (such as daily shopping and MRT TOD axis functionalities). However, due to legacy mixed industrial-residential development patterns, the per capita green space across a vast expanse of minimum statistical units falls far below the statutory scarcity threshold ($\leq 2 \text{ m}^2 / \text{person}$). Furthermore, the adjacent Zhongzheng Road traffic artery exhibits the highest accident density in the entire district. Therefore, if homebuyers choose to invest here, their site selection should precisely move

away from major longitudinal arteries and the Tandigou channel by a specified spatial buffer distance, alongside implementing standard indoor soundproofing and airtight hardware configurations.

Xinzhuang Sub-city Center Core Area (Modern Environmental Edge vs. Lagging Foundational Functionality): This sub-district easily surpasses World Health Organization (WHO) excellence benchmarks regarding urban planning public facility allocation, intact ecological green space, and active transportation network coverage. Nevertheless, the multi-criteria evaluation model exposes a lingering development and functional infill phase within its core interior blocks concerning the micro-accessibility of affordable daily convenience supermarkets and primary healthcare facilities. It is recommended that homebuyers adopt a spatial site-selection strategy of "purchasing homes along interior lines, but choosing commercial offices along major avenues."

6.3 Research Contributions and Practical Outlook of the Spatial Decision Model

This project successfully demonstrates how open-source geographic information systems (QGIS), spatial topology, and choropleth statistical tools can unify discrete urban public transit networks, point-based traffic accidents, and crowdsourced volunteered geographic information (VGI) into a micro-scale geometric carrier: the Minimum Statistical Unit (MSU). Moving beyond coarse metrics traditionally bounded by administrative villages, this study successfully constructs an urban living landscape analytical framework that bridges micro-scale geographical contexts with critical reflections on environmental injustice.

The finalized, scientifically quantified layout maps produced by this project do not merely provide a precision "urban acupuncture" framework for future sustainable and livable urban planning in Xinzhuang District; they deliver a highly innovative decision support model for rational private home purchases, social housing site-selection evaluations, and the human-centric remodeling of pedestrian environments.

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[\[https://data.ntpc.gov.tw/datasets/34b402a8-53d9-483d-9406-24a682c2d6dc\]](https://data.ntpc.gov.tw/datasets/34b402a8-53d9-483d-9406-24a682c2d6dc)

- 新北市政府資料開放平臺。新北市公共自行車租賃系統(YouBike2.0)。取自：

[\[https://data.ntpc.gov.tw/datasets/010e5b15-3823-4b20-b401-b1cf000550c5\]](https://data.ntpc.gov.tw/datasets/010e5b15-3823-4b20-b401-b1cf000550c5)

- 新北市政府資料開放平臺。新北市自行車道資料。取自：

[\[https://data.ntpc.gov.tw/datasets/996a600d-fd74-4a49-92d3-9464d2606e3f\]](https://data.ntpc.gov.tw/datasets/996a600d-fd74-4a49-92d3-9464d2606e3f)

- 水利空間資訊服務平台。河川流域範圍圖。取自：

[\[https://gic.wra.gov.tw/Gis/gic/API/Google/Index.aspx\]](https://gic.wra.gov.tw/Gis/gic/API/Google/Index.aspx)

- 新北市捷運工程局。3.1 車站規劃原則與設施需求。取自：

[\[https://www.dorts.ntpc.gov.tw/cdn/file/20220817/4c1951da-4fd9-4e2e-b17e-ef0f1c2d578e\]](https://www.dorts.ntpc.gov.tw/cdn/file/20220817/4c1951da-4fd9-4e2e-b17e-ef0f1c2d578e)

- 政府資料開放平台。114 年傷亡道路交通事故資料。取自：

[\[https://data.gov.tw/dataset/177136\]](https://data.gov.tw/dataset/177136)